Learning with Little Side Information

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Abstract

We address the problem of classification where, at test time, side information, a label from a different task, is available alongside the input features. Our objective is to design a classifier that can effectively leverage this additional side information. This side information cannot be treated as an additional feature because the training data set is assumed to be fragmented – we do not have access to simultaneous joint observations of labels and side information, together with the features. We develop an algorithm that operates under this setting by imputing the missing side information. Its behavior is governed by a regularization hyperparameter that determines the degree of trust placed in the imputed values. We derive an upper bound on the excess risk as a function of this hyperparameter. In the limit of infinitely many samples, the optimal regularization strategy becomes binary: one should either fully trust or entirely ignore the side information. Additionally, we establish a non-vanishing lower bound on the excess risk achievable by any algorithm, highlighting a fundamental limitation caused by the absence of joint label–side information observations during training.

1 Introduction

Consider the following scenario in university admissions. A university (the primary university) seeks to admit candidates with the right aptitude for its program. However, a candidate's aptitude is not directly observable and must instead be inferred from their profile, which includes exam scores, experience, and other attributes. Typically, admission decisions are based on historical data, profiles of past candidates and their subsequent performance in the program.

Now, suppose a candidate, in addition to their own profile, also discloses the outcome of their application at another university (the secondary university). This additional information, referred to as *side information*, raises a central question: How should the primary university leverage the decision of the secondary university to improve the accuracy of its own admission process?

The value of such side information depends crucially on the relation between the two universities. If the secondary university shares similar academic standards, subject focus, and reputation, the side information can be beneficial in assessing the candidate's aptitude. Conversely, if the two institutions differ significantly in their requirements or selectivity, the side information may be misleading and could result in suboptimal decisions. In general, the degree of alignment between the objectives and standards of the two universities determines how much weight the primary university should assign to the side information.

We consider the setting where, historically, no candidate has admission outcomes from both universities at the same time. Thus the past data includes rows with student features and admission decisions from either the primary university or the secondary university. Indeed if decisions of both universities were available, the secondary university decision could be absorbed readily as an additional feature in the learning problem. The data with the primary university decisions is also assumed to be more voluminous compared to that with secondary university decisions. Both these assumptions are motivated from real life experiences – students often apply to universities near their homes whereby past student data is often partitioned as above when the primary and secondary universities are geographically far apart. A university is also likely to have access to very limited data of other universities.

In this work, we formally study this problem and quantify how the error in admission decisions relates to the amount of trust placed in side information. We further characterize how much trust to place in side information, assuming the true relationship between candidate features, the primary university's decision, and the secondary university's decision is known a priori.

We formalize this problem within a classification framework. The primary and secondary university classifiers predict the admission outcome for a candidate, referred to as the label and side information, respectively. The primary university trains its classifier using both its own training data and the secondary university's decision rule. This allows the primary university to incorporate the secondary university's knowledge into its own inference process. At test time, when both the candidate's profile and the secondary university's admission decision are available, the primary university aims to construct a composite classifier that combines its own prediction with the side information, ensuring performance at least as good as the primary classifier alone.

Recall that the two training datasets are completely disjoint – referred to as a *partitioned dataset*. The challenge with a partitioned dataset is that very different underlying relationships between candidate features, the primary university's decision, and the secondary university's decision can produce the same observed data. More precisely, the *joint distribution* of features, label and side information is *unlearnable* from partitioned data. As a result, there is a fundamental limitation to the achievable accuracy of prediction.

A central contribution of this work is the design and analysis of an algorithm for combining the primary and secondary classifiers. The proposed algorithm first trains the secondary classifier and then uses it to impute the missing side information in the primary classifier's training dataset. However, the estimated side information may not be fully accurate, as its accuracy depends on both the size of the secondary classifier's training data and the complexity of the relationship between features and side information. Furthermore, since no historical data point contains both the true label and true side information (due to the partitioned dataset), any observed correlation between the label and the imputed side information could be spurious. To address this, we introduce a regularization mechanism that controls the influence of the estimated side information during training. The inverse of the regularization strength can be interpreted as a degree of trust in the imputed values. A high regularization value corresponds to little trust on the side information, while a low regularization value corresponds to greater trust. The optimal choice of this regularization parameter depends on knowledge of the true relationship between candidate aptitude, the label, and side information.

We establish a theoretical upper bound on the excess risk of the resulting composite classifier as a function of the regularization parameter. This upper bound exhibits the expected limiting behaviors: in the case of zero trust (infinite regularization), the bound reduces to that of the primary classifier alone; with complete trust (zero regularization), it becomes (roughly) the sum of the bounds for the primary and secondary classifiers. Importantly, our analysis shows that this upper bound on excess risk does not necessarily vanish, even with infinite training data. Indeed we confirm the non-vanishingness of excess risk by establishing a corresponding non-vanishing minimax lower bound on the excess risk. This fundamental limitation arises from the partitioned data setting, where features, labels, and side information are never jointly observed.

We do an ex-post analysis of the upper bound to find the optimal regularization assuming the data distribution is known. We find that the upper bound exhibits a bang-bang behavior – one must either fully trust or completely ignore the side information. Though this conclusion pertains only to the upper bound and may not reflect the behavior of the actual excess risk, our simulations for the actual excess risk suggest that a similar behavior is observed there too.

1.1 Related works

The use of extra information, in addition to the usual feature-label instances, has been studied in the learning theory literature. We highlight a few of those works and compare them with ours. To the best of our knowledge, our setting of using a partitioned dataset for training and having side information accessible at test time has not been previously studied.

Learning Using Privileged Information (LUPI), introduced by Vapnik & Vashist (2009), is a learning paradigm that leverages additional information available during training, but not at test time, to enhance

learning performance. This is analogous to a teacher who, in addition to providing answers, also offers hints or explanations to students. Subsequent works have extended the LUPI framework to various domains, including Vapnik et al. (2015); Sacco et al. (2021); Song et al. (2021); Wang et al. (2018). In contrast, our work considers a setting where side information is not observed jointly with labels during training (i.e., the dataset is partitioned), and becomes available only at test time, making it distinct from the above approaches.

Our problem can be viewed as learning from an incomplete training dataset—in our case, a partitioned dataset. A rich body of literature addresses missing data imputation, including Bertsimas et al. (2018); Van Buuren (2007); Ghahramani & Jordan (1993); Stekhoven & Bühlmann (2012); Nguyen et al. (2023). Unlike these works, which primarily focus on the soundness of imputation methods, our approach treats imputation as an intermediate step, with the main objective of analyzing the excess risk of the proposed algorithm. Additionally, we apply regularization to the imputed values based on our confidence in their accuracy, introducing an added layer of caution. This emphasis on excess risk analysis distinguishes our approach from existing imputation-focused literature.

Our work shares certain similarities with transfer learning, as discussed in Hosna et al. (2022), where knowledge from an auxiliary task is used to improve performance on a main task. However, in standard transfer learning, auxiliary task labels are typically available during both training and testing, and are jointly observed with the main task labels during training. In contrast, our setting involves side information that is not jointly observed with labels during training, making it structurally different from conventional transfer learning approaches.

Several other works, such as Adel & Levene (2023); Chen et al. (2012); Jonschkowski et al. (2015); Mollaysa et al. (2017), define and utilize side information in their own ways across different contexts. A common aspect of these approaches is that side information is jointly observed with labels in the training data and is typically not available at test time (with the exception of Adel & Levene (2023), where side information is also available at test time). In contrast, our setting differs in formulation: we consider a partitioned training dataset and assume that side information is available only at test time.

2 Problem Formulation

2.1 Notation Scheme

Random variables are denoted by capital letters (e.g., X), with their specific instances represented by the corresponding lowercase letters (e.g., x). The space over which a random variable is defined is denoted using the corresponding calligraphic letter (e.g., $\mathcal{X}$). Risks are represented by capital letters (e.g., R, L), while empirical risks are indicated by placing a hat over the corresponding letter (e.g., $\hat{R}, \hat{L}$). Model parameters, such as weights, are denoted by lowercase letters (e.g., w, u). The optimal weights minimizing the population risk are denoted with a superscript star (e.g., $w^\star$), and those minimizing the empirical risk are denoted with a hat (e.g., $\hat{w}$). We use sans-serif letters to denote important constants (w.r.t. this paper, e.g., W). For random variables X and Y, we write $\mathbb{E}_X[f(X,Y) \mid Y]$ simply as $\mathbb{E}_X[f(X,Y)]$ when the conditioning variable is clear from context.

2.2 Modeling

Let $\mathcal{X} \subset \mathbb{R}^d$ denote the feature space, $\mathcal{Y} = \{-1, 1\}$ the label space, and $\mathcal{Z} = \{-1, 1\}$ the side information space. Let (X, Y, Z) be a random variable representing the true data generating process over $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$. We assume the following properties:

$$
\begin{aligned}
&\textbf{(C1) Boundedness:}\ \exists \kappa > 0\ \ \|X\| \le \kappa \text{ almost surely,}\\
&\textbf{(C2) Regularity:}\ X \text{ and each of its marginals have a density function, and}\\
&\text{their cumulative distribution functions are Lipschitz continuous,}\\
&\textbf{(C3) Positive definiteness:}\ \mathbb{E}[(1,X)(1,X)^T] \succ 0,\\
&\textbf{(C4) Non-separability:}\ \text{For all } x, z,\\
&\quad \mathbb{P}(Y=1 \mid X=x, Z=z) \notin \{0,1\},\\
&\quad \mathbb{P}(Y=1 \mid X=x) \notin \{0,1\},\\
&\quad \mathbb{P}(Z=1 \mid X=x) \notin \{0,1\}.
\end{aligned}
\tag{1}
$$

Consider a dataset $\{(x_k, y_k, z_k)\}_{k=1}^{n+m}$ consisting of i.i.d. samples drawn from the joint distribution of (X, Y, Z). We partition this dataset as follows.

$$
\begin{aligned}
D &= \{(x_i, y_i)\}_{i=1}^{n} \quad (\text{missing } z_i),\\
S &= \{(x_j, z_j)\}_{j=n+1}^{n+m} \quad (\text{missing } y_j).
\end{aligned}
\tag{2}
$$

Here, D contains feature-label pairs with missing side information, and S contains feature–side information pairs with missing labels. At test time, a sample (x, y, z) is drawn from the joint distribution of (X, Y, Z). The objective is to learn a classifier using both S and D in order to accurately predict the target label y based on the features x and side information z.

We employ a logistic regression classifier with an intercept term and use a decision threshold of 0.5 for binary prediction. Define the sigmoid function,

$$
\sigma(t) := \frac{1}{1+\exp(-t)}, \quad t \in \mathbb{R}.
$$

and let the logistic loss for a feature vector $a \in \mathbb{R}^k$, label $b \in \{-1, 1\}$ and a weight vector $c \in \mathbb{R}^{k+1}$ be given by

$$
\ell(a, b; c) := -\ln(\sigma(b(c^\top(1, a)))) \tag{3}
$$

To characterize the excess risk of the hypothesis learned by our algorithm, we define the following weights and associated risk.

We now define three different classifiers. Let $u := (u_c, u_x) \in \mathbb{R}^{d+1}$ parameterize a classifier $\phi_u : \mathcal{X} \to \mathcal{Z}$ given by

$$
\phi_u(x) := 2\mathbb{1}_{\{\sigma(u_c + u_x^\top x) > 0.5\}} - 1.
$$

The logistic population risk of a classifier parameterised by u is defined as (with $a = X, b = Z, c = (u_c, u_x)$ in the logisictic loss)

$$
L(u) := \mathbb{E}_{(X,Y,Z)}[\ell(X, Z; u)] = \mathbb{E}_{(X,Z)}[-\ln(\sigma(Z(u_c + u_x^\top X)))]. \tag{4}
$$

The optimal weight $u^\star := (u_c^\star, u_x^\star)$ minimizes the population risk:

$$
u^\star := \arg\min_u L(u). \tag{5}
$$

From the non-separability assumption, we have $\|u^\star\| < \infty$ (a consequence of Lemma 6.5). The population risk under the zero-one loss is defined as

$$
L_{(0,1)}(u) := \mathbb{E}_{(X,Y,Z)}[\mathbb{1}_{\{Z(u_c + u_x^\top X) \le 0\}}] = \mathbb{E}_{(X,Z)}[\mathbb{1}_{\{Z(u_c + u_x^\top X) \le 0\}}]. \tag{6}
$$

The logistic empirical risk over S of a classifier parameterised by u is defined as (with $a = x_j, b = z_j, c = (u_c, u_x)$ in the logistic loss)

$$
\hat{L}(u) := \frac{1}{m} \sum_{(x_j, z_j) \in S} \ell(x_j, z_j; u) = \frac{1}{m} \sum_{(x_j, z_j) \in S} -\ln(\sigma(z_j(u_c + u_x^\top x_j))).
$$

For a regularization parameter $\mu \in \mathbb{R}$, the regularized empirical minimizer $\hat{u} := (\hat{u}_c, \hat{u}_x)$[1] is given by

$$\hat{u} := \arg\min_u \hat{L}(u) + \mu\|u\|^2. \tag{7}$$

The corresponding classifiers associated with $u^\star$ and $\hat{u}$ are denoted by

$$\phi^\star(x) := \phi_{u^\star}(x) \text{ and } \quad \hat{\phi}(x) := \phi_{\hat{u}}(x). \tag{8}$$

Let $v := (v_c, v_x) \in \mathbb{R}^{d+1}$ parameterize a classifier $\psi_v : \mathcal{X} \to \mathcal{Y}$ given by

$$\psi_v(x) := 2\mathbb{1}_{\{\sigma(v_c + v_x^\top x) > 0.5\}} - 1.$$

The logistic population risk of a classifier parameterised by v is defined as (with $a = X, b = Y, c = (v_c, v_x)$ in the logistic loss)

$$T(v) := \mathbb{E}_{(X,Y.Z)}[\ell(X, Y; v)] = \mathbb{E}_{(X,Y)}[-\ln(\sigma(Y(v_c + v_x^\top X)))].,$$

and the optimal weight $v^\star := (v_c^\star, v_x^\star)$ is given by

$$v^\star := \arg\min_v T(v). \tag{9}$$

Note that from the non-separability assumption, we have $\|v^\star\| < \infty$ (a consequence of Lemma 6.5). Finally let $w := (w_c, w_x, w_z) \in \mathbb{R}^{d+2}$ parameterize a classifier $\pi_w : \mathcal{X} \times \mathcal{Z} \to \mathcal{Y}$ given by

$$\pi_w(x, z) := 2\mathbb{1}_{\{\sigma(w_c + w_x^\top x + w_z z) > 0.5\}} - 1.$$

The logistic population risk of a classifier parameterised by w is defined as (with $a = (X, Z), b = Y, c = (w_c, w_x, w_z)$ in the logistic loss)

$$R(w) := \mathbb{E}_{(X,Y,Z)}[\ell(X, Z, Y; w)] = \mathbb{E}_{(X,Y,Z)}[-\ln(\sigma(Y(w_c + w_x^\top X + w_z Z)))]. \tag{10}$$

The optimal weight $w^\star := (w_c^\star, w_x^\star, w_z^\star)$ is defined as

$$w^\star := \arg\min_w R(w). \tag{11}$$

Note that from the non-separability assumption, $\|w^\star\| < \infty$ (a consequence of Lemma 6.5). For a regularization parameter $\lambda := (\lambda_c, \lambda_x, \lambda_z) \in \mathbb{R}^3$, the regularized logisitic population risk is

$$R_\lambda(w) := R(w) + \lambda_c|w_c|^2 + \lambda_x\|w_x\|^2 + \lambda_z|w_z|^2,$$

with minimizer

$$w_\lambda^\star := \arg\min_{w \in \mathbb{R}^{d+2}} R_\lambda(w). \tag{12}$$

Given a mapping $\phi : \mathcal{X} \to \mathcal{Z}$, the *logisitic ϕ-population risk* over (X, Y) is

$$R^\phi(w) := \mathbb{E}_{(X,Y)}[\ell(X, \phi(X), Y; w)].$$

The corresponding regularized risk and its minimizer are

$$R_\lambda^\phi(w) := R^\phi(w) + \lambda_c|w_c|^2 + \lambda_x\|w_x\|^2 + \lambda_z|w_z|^2, \tag{13}$$

and

$$w_{\phi,\lambda}^\star := \arg\min_w R_\lambda^\phi(w) \tag{14}$$

respectively.

[1]Note that the notation $\hat{u}$ supresses the implicit depende on the regulariser μ for brevity.

The logisitic empirical risk over D with imputed side information $\hat{\phi}(x_i)$ described in (8) for a classifier parameterised by w is defined as (with $a = (x_i, \hat{\phi}(x_i)), b = y_i, c = (w_c, w_x, w_z)$ in the logistic loss)

$$\hat{R}^{\hat{\phi}}(w) := \frac{1}{n} \sum_{(x_i,y_i)\in D} \ell(x_i, \hat{\phi}(x_i), y_i; w) = \frac{1}{n} \sum_{(x_i,y_i)\in D} -\ln(\sigma(y_i(w_c + w_x^\top x_i + w_z \hat{\phi}(x_i)))).$$

The corresponding regularized empirical minimizer is

$$\hat{w}_\lambda := \arg\min_w \left[\hat{R}^{\hat{\phi}}(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2 \right]. \tag{15}$$

Our goal is to analyze the excess risk of the learned classifier $\hat{w}_\lambda$ in terms of the various population-optimal weights described earlier. [2]

3 Algorithm

We present our learning algorithm for leveraging side information in classification problems with partitioned datasets. The procedure consists of two main stages:

1. Learning the Feature to Side Information mapping: Select a regularization parameter μ and learn a classifier $\hat{u} \equiv \hat{\phi} : \mathcal{X} \to \mathcal{Z}$ from the sample S using logistic regression (thresholded at 0.5). This classifier is used to impute the missing Z-values in the primary dataset D.

2. Learning the Feature and inputed Side Information to Label mapping with regularisation: Choose regularization parameters $(\lambda_c, \lambda_x, \lambda_z)$. For each $(x_i, y_i) \in D$, impute $z_i = \hat{\phi}(x_i)$. Train a regularized logistic regression model (thresholded at 0.5) to learn a classifier $\hat{w}_\lambda : \mathcal{X} \times \mathcal{Z} \to \mathcal{Y}$.

Among all regularization choices, only the selection of λ_z is non-trivial and thus becomes the primary focus of both theoretical and empirical analysis in this work.

Data: $D = \{(x_i, y_i)\}_{i=1}^n$, $S = \{(x_j, z_j)\}_{j=1}^m$
Result: Hypothesis $\hat{w}_\lambda$

`// Step 1: Select the optimal regularization parameter` $\mu = \frac{c}{\sqrt{m}}$ `through validation, where` c `is a constant chosen to minimize side information prediction error on the validation set.`

`// Step 2: Learn` $\hat{u}$ `by minimizing the regularized empirical risk over` S`:`

$$\hat{u} := \arg\min_{u\in\mathbb{R}^{d+1}} \hat{L}(u) + \mu\|u\|^2$$

`// Step 3: For each` $(x_i, y_i) \in D$`, impute` $z_i = \hat{\phi}(x_i)$`.`

`// Step 4: Select regularization parameters` $(\lambda_c, \lambda_x, \lambda_z)$ `for the primary classifier, where` $\lambda_c = \lambda_x = \frac{1}{\sqrt{n}}$`.`

`// Step 5: Learn` $\hat{w}_\lambda$ `by minimizing the regularized empirical risk over` D `with imputed` z_i`:`

$$\hat{w}_\lambda := \arg\min_{w\in\mathbb{R}^{d+2}} \hat{R}^{\hat{\phi}}(w) + \lambda_c |w_c|^2 + \lambda_x \|w_x\|^2 + \lambda_z |w_z|^2.$$

`// Step 6: At test time, given` (x, z) `(where` z `is observed and not imputed from` x `using the side information model), predict` $\hat{y} = 1$ `if` $\hat{w}_{c,\lambda} + \hat{w}_{\lambda,x}^\top x + \hat{w}_{\lambda,z} z \geq 0.5$`, else` $\hat{y} = -1$`.`

Algorithm 1: Learning with Side Information

[2] Note that the notation $\hat{w}_\lambda$ suppresses the implicit dependence on $\hat{\phi}$ for brevity.

4 Analysis of the Algorithm

In this section, we analyze the excess risk of Algorithm 1. Our main theoretical result, Theorem 4.1, provides an explicit upper bound on the excess risk as a function of the sample sizes, distribution-dependent quantities, and the regularization parameter. This bound characterizes how the choice of regularizer mediates the trade-off between relying on imputed side information and using only the primary features. Corollaries 4.1.1, 4.1.2, and 4.1.3 further offers guidance for selecting the regularization parameter in different regimes.

In addition, Theorem 4.2 establishes a minimax lower bound on the excess risk for any algorithm, quantifying the intrinsic difficulty of learning from partitioned datasets in which features, labels, and side information are never observed jointly. Finally, we conduct simulations of the excess risk on a representative distribution to empirically investigate the choice of the optimal regularization parameter.

4.1 Main Result: Upper Bound on Excess Risk

We now state our main theorem, which provides an explicit upper bound on the excess risk as a function of the sample sizes m and n, relevant distribution-dependent quantities, and the choice of regularizer λ_z.

Let

$$\mathsf{W} := \max\{\sup_{\lambda} \|w^\star_{\phi^\star,\lambda}\|, \sup_{\lambda} \|\hat{w}_\lambda\|, \sup_{\lambda} \|w^\star_\lambda\|, \|w^\star\|, \|u^\star\|, \|\hat{u}\|\}. \tag{16}$$

We show in Proposition 1 that with high probability, $\mathsf{W} < \infty$.

Theorem 4.1. *Let $u^\star$ and $v^\star$ be as defined in (5) and (9), respectively. Let $w^\star$, $w^\star_\lambda$, and W be as defined in (11), (12) and (16) respetively. Consider the partitioned training datasets $D = \{(x_i, y_i)\}_{i=1}^n$ and $S = \{(x_j, z_j)\}_{j=1}^m$, as defined in (2), and let $\delta > 0$. Take the regularizer λ as*

$$\lambda = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \lambda_z\right) \in \mathbb{R}^3,$$

where $\lambda_z \geq \frac{1}{\sqrt{n}}$, controls the regularization strength for the side information component. Then, with probability at least $1 - 4\delta$, the following inequality holds.

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} &+ \min\left\{\frac{\lambda_z |w^\star_z|^2}{T(v^\star) - R(w^\star)}, 1 + \frac{\lambda_z |w^\star_{\lambda,z}|^2}{T(v^\star) - R(w^\star)}\right\}\Big(T(v^\star) - R(w^\star)\Big) \\ &+ \min\left\{\frac{\ln(2)}{\lambda_z}, \mathsf{W}\right\} \ln(4)\left(\frac{C_2}{\sqrt{m}} + L(u^\star)\right), \end{aligned} \tag{17}$$

where $C_1 = \mathcal{O}\left(W^2 + \kappa W\sqrt{\ln(1/\delta)} + \kappa^2 \ln(1/\delta)\right)$ and $C_2 = \mathcal{O}\left(\|u^*\|\sqrt{\ln(1/\delta)}\right)$ are constants.[3]

Proof Outline. To analyze the excess risk, we decompose the left-hand side of (17) into four terms, each capturing a distinct source of error:

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^\star) = &\underbrace{R(\hat{w}_\lambda) - R^{\hat{\phi}}(\hat{w}_\lambda)}_{\text{(A)}} + \underbrace{R^{\hat{\phi}}(\hat{w}_\lambda) - R^{\hat{\phi}}(w^\star_\lambda)}_{\text{(B)}} \\ &+ \underbrace{R^{\hat{\phi}}(w^\star_\lambda) - R(w^\star_\lambda)}_{\text{(C)}} + \underbrace{R(w^\star_\lambda) - R(w^\star)}_{\text{(D)}}. \end{aligned}$$

The first and third terms, (A) and (C), quantify the estimation error resulting from the use of the imputed side information $\hat{\phi}$. This error depends on the complexity of the relationship between (X, Z) quantified by $L(u^\star)$, the sample size m of the side information dataset S, and the regularization parameter λ_z, which

[3]The consequence of the assmption $m < n$ is that the term $1/\sqrt{m}$ on the RHS does not get absorbed in the $1/\sqrt{n}$.

determines the extent to which the model relies on the imputed side information. These terms contribute the following to the upper bound:

$$\min\left(\frac{\ln(2)}{\lambda_z},\ \mathsf{W}\right) \times \ln(4)\left(\frac{C_2}{\sqrt{m}} + L(u^\star)\right).$$

The second term, (B), corresponds to the estimation error in the standard supervised learning setting for the primary classifier, which results in the contribution $\frac{C_1}{\sqrt{n}}$ to the upper bound in (17).

Finally, the fourth term, (D), captures the gap between the risk of the optimal regularized classifier and the risk of the optimal classifier. This term contributes

$$\min\left\{\frac{\lambda_z|w_z^\star|^2}{T(v^\star)-R(w^\star)},\ 1+\frac{\lambda_z|w_{\lambda_z}^\star|^2}{T(v^\star)-R(w^\star)}\right\}(T(v^\star)-R(w^\star))$$

to the upper bound in (17). Note that $v^\star$ coincides with $w_\lambda^\star$ when $\lambda_z = \infty$, so this term quantifies the proportion of the worst-case gap $T(v^\star) - R(w^\star)$ (with the worst at $\lambda_z = \infty$) that is realized, depending on the choice of the regularizer λ_z. □

4.1.1 Interpretation of the Excess Risk Bound

To provide intuition for the upper bound in Theorem 4.1, consider the university admission scenario. Here, the feature $x \in \mathbb{R}^d$ represents a candidate's academic scores, the side information z is the admission decision of a secondary university, and the label y is the decision of the primary university. The excess risk $R(\hat{w}_\lambda) - R(w^*)$ represents the performance gap between our learned classifier $\hat{w}_\lambda$ and the benchmark optimal classifier w^*, which has access to the true joint distribution of (X, Y, Z). This risk is controlled by the regularization parameter λ_z and consists of three distinct components.

The first term, $\mathcal{O}(1/\sqrt{n})$, is the standard estimation error associated with the sample size n of the primary university's dataset. This error is unavoidable and decays as more primary data becomes available.

The second term captures the potential gain from incorporating z. It is proportional to the gap $T(v^*) - R(w^*)$, which measures how much knowing the true side information improves prediction accuracy over using features x alone. This term is controlled linearly by λ_z. If z carries meaningful information (i.e., the gap is significant), the bound is lowered by reducing λ_z, which encourages the model to assign larger weight to the side information.

The third term quantifies the error introduced by imputing z from x using the secondary university sample. It depends on the sample size m of the secondary university's data and the optimal prediction error $L(u^*)$, which reflects the complexity of the relationship between features and side information. This term is inversely controlled by λ_z (scaling as $1/\lambda_z$). If the features cannot accurately predict z, the imputed values are noisy, and a large λ_z is required to ignore the unreliable imputed data.

The interplay between the second and third terms creates a tension in selecting λ_z. If z is highly informative for predicting y (large gap $T(v^*) - R(w^*)$), a smaller λ_z is desirable to leverage this information. However, if x poorly predicts z (large $L(u^*)$), a larger λ_z is necessary to mitigate the noise from imputed side information. The regularizer λ_z must balance these two competing terms. In this case, the optimal λ_z must find a balance to extract the signal in z while damping the noise from imperfect imputation. In most other cases, where z is either useless or perfectly predictable, the terms do not conflict, and the choice of λ_z becomes straightforward. As we will show in subsequent sections, despite this tension, the optimal λ_z often exhibits bang-bang behavior, taking on extreme values (either very large or very small) depending on which factor is more limiting: the noise introduced by imperfectly imputed z, or the dependence of y on z. This distinction is determined by the underlying distributional properties.

4.1.2 Retrospective Regularizer Selection

Theorem 4.1 provides an explicit upper bound on the excess risk as a function of the regularization parameter λ_z. This upper bound holds for all values of λ_z and all distributions. We are interested in determining the

value of λ_z that minimizes this upper bound. Since λ_z is a hyperparameter, its optimal value depends on the underlying population distribution. Under certain assumptions, we show that the choice for λ_z may be restricted to either 0 or ∞. To facilitate this analysis, we work with a relaxed version of the upper bound (17) from Theorem 4.1. Specifically, by applying Lemma 8.2 (given in the Appendix), we have $\lambda_z|w^\star_{\lambda,z}|^2 \leq \ln(2)$, so we obtain the following relaxed bound.

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} &+ \min\left\{\frac{\lambda_z|w^\star_z|^2}{T(v^\star)-R(w^\star)}, 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}\right\}\Big(T(v^\star)-R(w^\star)\Big) \\ &+ \min\left\{\frac{\ln(2)}{\lambda_z}, \mathsf{W}\right\}\ln(4)\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big), \end{aligned} \tag{18}$$

where $C_1 = \mathcal{O}\left(W^2 + \kappa W\sqrt{\ln(1/\delta)} + \kappa^2\ln(1/\delta)\right)$ and $C_2 = \mathcal{O}\left(\|u^*\|\sqrt{\ln(1/\delta)}\right)$ are constants. This form makes the λ_z dependence on excess risk explicit, enabling a more tractable analysis of its optimal value.

Let

$$\begin{aligned} \Lambda^\star_z := \arg\min_{\lambda_z} \frac{C_1}{\sqrt{n}} &+ \min\left\{\frac{\lambda_z|w^\star_z|^2}{T(v^\star)-R(w^\star)}, 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}\right\}\Big(T(v^\star)-R(w^\star)\Big) \\ &+ \min\left\{\frac{\ln(2)}{\lambda_z}, \mathsf{W}\right\}\ln(4)\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big), \end{aligned}$$

be the optimal (population distribution dependent) regularizer that minimizes the upper bound on the excess risk. For a given value of $\Lambda^\star_z$, three disjoint and exhaustive cases arise, each corresponding to a different regime for the optimal regularizer:

- **Case (a):**
$$\frac{\Lambda^\star_z|w^\star_z|^2}{T(v^\star)-R(w^\star)} \geq 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}.$$
This is the regime considered in Corollary 4.1.1.

- **Case (b):**
$$\frac{\Lambda^\star_z|w^\star_z|^2}{T(v^\star)-R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}.$$
and
$$\frac{\ln(2)}{\Lambda^\star_z} \geq \mathsf{W}.$$
This is the regime considered in Corollary 4.1.2.

- **Case (c):**
$$\frac{\Lambda^\star_z|w^\star_z|^2}{T(v^\star)-R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}.$$
and
$$\frac{\ln(2)}{\Lambda^\star_z} < \mathsf{W}.$$
This is the regime considered in Corollary 4.1.3.

Each case is formalized in the following corollaries, which provide guidance for selecting the regularization parameter in different regimes.

The first corollary shows that if $\Lambda^\star_z$ is sufficiently large, then it is $\Lambda^\star_z = \infty$.

Corollary 4.1.1 (Regime I: Large Regularizer)**.** *Suppose*

$$\frac{\Lambda^\star_z|w^\star_z|^2}{T(v^\star)-R(w^\star)} \geq 1 + \frac{\ln(2)}{T(v^\star)-R(w^\star)}.$$

Then $\Lambda_z^\star = \infty$ *and*

$$R(\hat{w}_\lambda) - R(w^\star) = \frac{C_1}{\sqrt{n}} + T(v^\star) - R(w^\star).$$

Proof. Consider the condition

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} \geq 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)}.$$

Under this condition, the minimum in the second term of the upper bound in (18) is achieved by the second argument.

Now consider λ_z such that $\lambda_z > \Lambda_z^\star$. This choice does not increase the contribution of the second term, as

$$\frac{\lambda_z |w_z^\star|^2}{T(v^\star) - R(w^\star)} \geq \frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} \geq 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)},$$

but it decreases the contribution of the third term in (18) to 0, as the first argument of the third term goes to zero. This decreases the value of the overall upper bound.

However, since $\Lambda_z^\star$ is assumed to be optimal, increasing λ_z beyond it should not reduce the upper bound. Thus, $\Lambda_z^\star = \infty$. Substituting this value back into (17) and noting that $\lambda_z |w_{\lambda,z}^\star|^2 \to 0$ as $\lambda_z \to \infty$, we obtain

$$R(\hat{w}_\lambda) - R(w^\star) = \frac{C_1}{\sqrt{n}} + T(v^\star) - R(w^\star).$$

□

The second corollary establishes that if $\Lambda_z^\star$ is sufficiently small, then the optimal value is $\Lambda_z^\star = \frac{1}{\sqrt{n}}$.

Corollary 4.1.2 (Regime II: Small Regularizer)**.** *Suppose*

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)}.$$

and

$$\frac{\ln(2)}{\Lambda_z^\star} \geq \mathsf{W}.$$

Then $\Lambda_z^\star = \frac{1}{\sqrt{n}}$ *and*

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + \mathsf{W} \ln(4) \Big(\frac{C_2}{\sqrt{m}} + L(u^\star) \Big).$$

Proof. Consider the condition

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)}.$$

and

$$\frac{\ln(2)}{\Lambda_z^\star} \geq \mathsf{W}.$$

Under these conditions, the minimum in the second term and the third term of the upper bound in (18) is achieved by the first argument, and the second argument respectively.

Let λ_z be such that $\lambda_z < \Lambda_z^\star$. This choice does not increase the contribution of the third term, as

$$\frac{\ln(2)}{\lambda_z} \geq \frac{\ln(2)}{\Lambda_z^\star} \geq \mathsf{W},$$

but it decreases the contribution of the second term as

$$\frac{\lambda_z |w_z^\star|^2}{T(v^\star) - R(w^\star)} < \frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)}.$$

This decreases the value of the overall upper bound.

However, since $\Lambda_z^\star$ is assumed to be optimal, decreasing λ_z below it should not reduce the upper bound. Thus, $\Lambda_z^\star = \frac{1}{\sqrt{n}}$, the minimum allowed value of λ_z. Substituting this value back into (17), we obtain

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + \mathsf{W}\ln(4)\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big).$$

□

The third corollary shows that if $\Lambda_z^\star$ is neither a big value nor a small value, then it scales with the size of side information dataset S as $\Lambda_z^\star = \mathcal{O}(m^{-1/4})$.

Corollary 4.1.3 (Regime III: Intermediate Regularizer)**.** *Suppose*

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)} \quad \textit{and}$$
$$\frac{\ln(2)}{\Lambda_z^\star} < \mathsf{W}.$$

Then the optimal regularizer satisfies

$$\Lambda_z^\star = \frac{\sqrt{2(\ln(2))^2\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big)}}{|w_z^\star|},$$

and the excess risk is bounded by

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + 2|w_z^\star|\sqrt{2(\ln(2))^2\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big)}.$$

Proof. Suppose the following two conditions hold:

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} \leq 1 + \frac{\Lambda_z^\star |w_{\lambda,z}^\star|^2}{T(v^\star) - R(w^\star)},$$
$$\frac{\ln(2)}{\Lambda_z^\star} \leq \mathsf{W}.$$

Under these conditions, the minimum in both the second and third terms of the upper bound in (18) is achieved by the first argument. Since the upper bound is minimized at $\Lambda_z^\star$, the derivative with respect to λ_z must vanish at this point. This gives

$$\frac{|w_z^\star|^2}{T(v^\star) - R(w^\star)} - \frac{\ln(2)}{\lambda_z^2}\ln(4)\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big)\Bigg|_{\lambda_z = \Lambda_z^\star} = 0.$$

Solving this yields

$$\Lambda_z^\star = \frac{\sqrt{2(\ln(2))^2\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big)}}{|w_z^\star|},$$

Substituting this value back into (17) gives

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + 2|w_z^\star|\sqrt{2(\ln(2))^2\left(\frac{C_2}{\sqrt{m}} + L(u^\star)\right)}.$$

□

We now show that under some suitable assumption, the Regime III considered in Corollary 4.1.3 does not arise.

Corollary 4.1.4. *Let n and m be sufficiently large, and let $L(u^\star) \leq 1$. Then the optimal value of $\Lambda_z^\star$ is either ∞ or 0.*

Proof. If

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} \geq 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)},$$

then from Corollary 4.1.1, we have $\Lambda_z^\star = \infty$. If

$$\frac{\Lambda_z^\star |w_z^\star|^2}{T(v^\star) - R(w^\star)} < 1 + \frac{\ln(2)}{T(v^\star) - R(w^\star)},$$

then, according to Corollaries 4.1.2 and 4.1.3, there are two possible values for the optimal regularizer. If $\Lambda_z^\star = \frac{1}{\sqrt{n}} \to 0$ (Corollary 4.1.2), the upper bound on the excess risk is

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + \mathsf{W} \ln(4) \Big(\frac{C_2}{\sqrt{m}} + L(u^\star) \Big) \tag{19}$$

and if $\Lambda_z^\star = \frac{\sqrt{2(\ln(2))^2 \left(\frac{C_2}{\sqrt{m}} + L(u^\star) \right)}}{|w_z^\star|}$ (Corollary 4.1.3), the upper bound on the excess risk is

$$R(\hat{w}_\lambda) - R(w^\star) \leq \frac{C_1}{\sqrt{n}} + 2|w_z^\star| \sqrt{2(\ln(2))^2 \Big(\frac{C_2}{\sqrt{m}} + L(u^\star) \Big)}. \tag{20}$$

The bound in (19) scales as $\mathcal{O}\left(\frac{1}{\sqrt{m}} + \ln(u^\star) \right)$, whereas the bound in (20) scales as $\mathcal{O}\left(\frac{1}{m^{1/4}} + \sqrt{\ln(u^\star)} \right)$. Thus, for sufficiently large n and m and $L(u^\star) \leq 1$, the bound in (19) is smaller, implying Regime III considered in Corollary 4.1.3 will never occur, and hence $\Lambda_z^\star = 0$.

□

4.2 Lower Bounds

In this section, we establish a fundamental limitation of learning from partitioned datasets. Specifically, we show that for any learning algorithm A, there exists a probability distribution of (X, Y, Z) such that if (D, S) is a partitioned data set defined in (2), the excess risk $R(A(D, S)) - R(w^\star)$ is bounded away from zero with non-vanishing probability. This demonstrates that, even with unlimited data, the inability to observe features, labels, and side information jointly imposes an irreducible barrier to learning performance.

Theorem 4.2 (Lower Bound on Excess Risk)**.** *Let (D, S) be a partitioned dataset defined in* (2)*, and let $(D, S) \mapsto A(D, S) \subset \mathbb{R}^{d+2}$ be any learning algorithm. Then there exists a distribution of (X, Y, Z) generating the partioned training datasets (D, S) such that for $w^\star$ defined in* (11)*, we have*

$$\mathbb{P}_{(X,Y,Z)} \left[R(A(D, S)) - R(w^\star) > 0.19 \right] \geq \frac{1}{2}. \tag{21}$$

Proof Outline. We construct two joint distributions, (X_1, Y_1, Z_1) and (X_2, Y_2, Z_2), such that the marginals (X_1, Y_1) and (X_1, Z_1) are identical to (X_2, Y_2) and (X_2, Z_2), respectively, but the joint distributions over (X, Y, Z) differ. For any true distribution (X_k, Y_k, Z_k) ($k \in \{1, 2\}$) generating the partitioned training data, the classification problem reduces to identifying which distribution generated the data. However, since the observed marginals are identical, no algorithm can distinguish between the two distributions based only on the partitioned datasets. Consequently, for any algorithm A, we can adversarially select the true distribution such that the excess risk is at least 0.19 with probability at least $\frac{1}{2}$. This establishes a non-vanishing lower bound on the excess risk for any learning strategy in the partitioned setting. □

4.3 Simulation Results

Corollaries 4.1.1, 4.1.2, and 4.1.3 offer theoretical guidance on the optimal choice of λ_z for minimizing the upper bound on excess risk. However, these results do not guarantee that the same value of λ_z is optimal for the true excess risk. To explore this, we conduct simulations to empirically identify the value of λ_z that minimizes the actual excess risk.

We conduct a simulation study to find the empirical distribution of the optimal regularization parameter, λ_z. We generate 100 different distributions over (X, Y, Z), each constructed as follows:

1. The features X are sampled from a Gaussain mixture model, whose parameters and mixing probabilities are randomly choosen.
2. Then the weight vectors u and w are sampled from standard normal distribution. Given fixed weights u, the binary variables Z is sampled via logistic models:

$$Z \sim \sigma((1, X)^\top u).$$

   Once we get Z, then for a fixed weight w, the binary variable Y is sample via logistic model:

$$Y \sim \sigma((1, X, Z)^\top w).$$

This process is repeated 100 times, each with a new random choice of parameters, resulting in a diverse family of distributions for evaluating the bang-bang behavior of the optimal λ_z.

For each distribution, we generate $m = 1000$ samples for the (X, Z) dataset (used for learning $X \to Z$), $n = 4000$ samples for the (X, Y) dataset (used for learning $X \times Z \to Y$), and $t = 20000$ samples for the test set (X, Y, Z) to compute the excess risk. For each value of λ_z in the candidate set

$$\{0.01, 0.1, 1.0, 10.0, 100.0, 1000.0, 10000.0, 100000.0\},$$

a model is trained and evaluated. The optimal λ_z for each distribtution is chosen as the value that yields the lowest empirical excess risk on the test set.

The frequency of each candidate λ value being chosen as the optimal λ_z across the 100 distributions is shown in Figure 1. The simulation results, summarized in the frequency plot, show a strong preference for the extreme values of the regularization parameter λ_z. This observed bang-bang behavior of λ_z provides empirical evidence suggesting a bang-bang nature of λ_z.

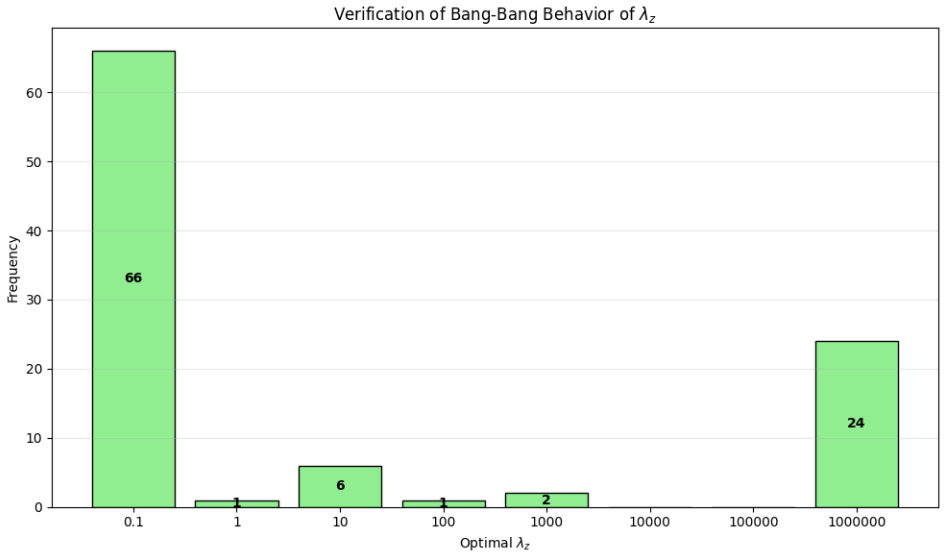

Figure 1: Frequency of optimal λ_z values across 100 independent simulation seeds. For each seed, λ_z is the value from the set $\{0.1, 1, 10, 100, 1000, 10000, 100000, 1000000\}$ that minimizes the empirical excess risk.

5 Conclusion

In this work, we addressed the problem of utilizing test-time side information for classification when joint observations of side information and labels were absent from the training dataset. We proposed an algorithm that imputes the missing side information during training. The algorithm's behavior is governed by a regularization hyperparameter, which quantifies the trust placed in the imputed data. We derived an upper bound on the excess risk as a function of this hyperparameter, noting that the optimal value depends on prior knowledge of the relationship between side information and labels. In the infinite-sample limit and some additional assumption, the optimal choice becomes binary: either fully integrate the imputed side information or completely ignore it. Furthermore, we established a fundamental limitation of this problem: the excess risk of any learning algorithm is lower-bounded by a non-vanishing term. This limitation arises directly from the lack of joint side information–label observations during training. A promising future direction is to explore the strategic aspect of this problem, where an agent decides whether to reveal side information to the classifier at test time.

References

Tameem Adel and Mark Levene. A general model for side information in neural networks. *Algorithms*, 16 (11):526, 2023.

Dimitris Bertsimas, Colin Pawlowski, and Ying Daisy Zhuo. From predictive methods to missing data imputation: an optimization approach. *Journal of Machine Learning Research*, 18(196):1–39, 2018.

Jixu Chen, Xiaoming Liu, and Siwei Lyu. Boosting with side information. In *Asian Conference on Computer Vision*, pp. 563–577. Springer, 2012.

Andrea Della Vecchia, Jaouad Mourtada, Ernesto De Vito, and Lorenzo Rosasco. Regularized erm on random subspaces. In *International Conference on Artificial Intelligence and Statistics*, pp. 4006–4014. PMLR, 2021.

Zoubin Ghahramani and Michael Jordan. Supervised learning from incomplete data via an em approach. *Advances in neural information processing systems*, 6, 1993.

Asmaul Hosna, Ethel Merry, Jigmey Gyalmo, Zulfikar Alom, Zeyar Aung, and Mohammad Abdul Azim. Transfer learning: a friendly introduction. *Journal of Big Data*, 9(1):102, 2022.

Rico Jonschkowski, Sebastian Höfer, and Oliver Brock. Patterns for learning with side information. *arXiv preprint arXiv:1511.06429*, 2015.

Amina Mollaysa, Pablo Strasser, and Alexandros Kalousis. Regularising non-linear models using feature side-information. In *International Conference on Machine Learning*, pp. 2508–2517. PMLR, 2017.

Thu Nguyen, Tuan L Vo, Pål Halvorsen, and Michael A Riegler. Imputation using training labels and classification via label imputation. *arXiv preprint arXiv:2311.16877*, 2023.

Ludovica Sacco, Dino Ienco, and Roberto Interdonato. A neural network strategy for supervised classification via the learning under privileged information paradigm. In *SEBD*, pp. 83–93, 2021.

Yuru Song, Zan Lou, Shan You, Erkun Yang, Fei Wang, Chen Qian, Changshui Zhang, and Xiaogang Wang. Learning with privileged tasks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 10685–10694, 2021.

Daniel J Stekhoven and Peter Bühlmann. Missforest—non-parametric missing value imputation for mixed-type data. *Bioinformatics*, 28(1):112–118, 2012.

Stef Van Buuren. Multiple imputation of discrete and continuous data by fully conditional specification. *Statistical methods in medical research*, 16(3):219–242, 2007.

Vladimir Vapnik and Akshay Vashist. A new learning paradigm: Learning using privileged information. *Neural networks*, 22(5-6):544–557, 2009.

Vladimir Vapnik, Rauf Izmailov, et al. Learning using privileged information: similarity control and knowledge transfer. *J. Mach. Learn. Res.*, 16(1):2023–2049, 2015.

Shangfei Wang, Shiyu Chen, Tanfang Chen, and Xiaoxiao Shi. Learning with privileged information for multi-label classification. *Pattern Recognition*, 81:60–70, 2018.

6 Appendix I: Lemmas About Generalization Gap from Existing Literature

In this section, we review key results from Della Vecchia et al. (2021) that will be used in the proof of the main theorem (Theorem 4.1). For convenience, we restate these theorems below. Before doing so, we summarize the assumptions and notation on which those results rely.

Let $\mathcal{H}$ be the hypothesis space, $\mathcal{H}^\star$ (the dual space of $\mathcal{H}$) be the feature space, and $\mathcal{B}$ the label space. Let $\mathtt{l}(b, r) \in \mathbb{R}$ denote the loss function, where $b \in \mathcal{B}$ and r is the score given by the inner product between the hypothesis and the feature vector. Consider the following assumptions.

(A1) Let $\mathcal{H}$ be a real separable Hilbert space with inner product $\langle \cdot, \cdot \rangle$, and let $\mathcal{B}$ be a Polish space. Let (A, B) be a random variable taking values in $\mathcal{H} \times \mathcal{B}$ with distribution P, and suppose there exists $\alpha > 0$ such that $\|A\| \leq \alpha$ almost surely.

(A2) The loss function $\mathtt{l} : \mathcal{B} \times \mathbb{R} \to [0, \infty)$ is Lipshitz in its second argument, namely their exists $G > 0$ such that for all $b \in \mathcal{B}$ and $r, r' \in \mathbb{R}$,

$$|\mathtt{l}(b, r) - \mathtt{l}(b, r')| \leq G|r - r'|.$$

(A3) Let $\mathfrak{L}(\mathtt{w}) := \int_{\mathcal{H}\times\mathcal{B}} \mathtt{l}(b, \langle \mathtt{w}, a\rangle)\, dP(a, b)$ denote the population risk. Suppose there exists $\mathtt{w}_* \in \mathcal{H}$ such that $\mathfrak{L}(\mathtt{w}_*) = \sup_{\mathtt{w}\in\mathcal{H}} \mathfrak{L}(\mathtt{w})$. For $\lambda > 0$, let $\widehat{\mathtt{w}}_\lambda \in \mathcal{H}$ be defined by $\widehat{\mathtt{w}}_\lambda := \arg\min_{\mathtt{w}\in\mathcal{H}} \frac{1}{n} \sum_{i=1}^n \mathtt{l}(b_i, \langle \mathtt{w}, a_i\rangle) + \lambda\|\mathtt{w}\|^2$.

We first restate the Theorem 1 in Della Vecchia et al. (2021).

Lemma 6.1 (Theorem 1, Della Vecchia et al. (2021))**.** *Suppose the assumptions (A1), (A2), and (A3) hold. Then for sample size n and for all $\lambda > 0$ and $0 < \delta < 1$, with probability at least $1-\delta$, the following inequality holds.*

$$\mathfrak{L}(\widehat{\mathtt{w}}_\lambda) - \mathfrak{L}(\mathtt{w}_*) \leq \frac{G^2\alpha^2 \ln(1/\delta)}{\lambda n} + \lambda\|\mathtt{w}_*\|^2.$$

Hence letting $\lambda = (G\alpha/\|\mathtt{w}_*\|)\sqrt{\ln(1/\delta)/n}$ leads to a rate of $\mathcal{O}\Big(\|\mathtt{w}_*\|\sqrt{\frac{\ln(1/\delta)}{n}}\Big)$.

We now use this theorem to establish an upper bound on the excess risk between $\hat{u}$ (which paramterizes $\hat{\phi}$) and $u^\star$ (which paramterizes $\phi^\star$).

Lemma 6.2. *Let $\hat{u}$ and $u^\star$ be defined in (5) and (7), respectively. Let $S = \{(x_j, z_j)\}_{j=n+1}^{n+m}$ and $\delta \in (0, 1)$. Let $\mu = \frac{1}{\|u^\star\|}\sqrt{\frac{\ln(1/\delta)}{m}}$. Then, with probability at least $1 - \delta$, we have*

$$L(\hat{u}) - L(u^\star) \leq \epsilon_{\mathtt{u}}, \tag{22}$$

where $\epsilon_{\mathtt{u}} = \mathcal{O}\Big(\|u^\star\|\sqrt{\frac{\ln(1/\delta)}{m}}\Big)$.

Proof. Apply Theorem 1 in Della Vecchia et al. (2021) with $G = 1$, $\mathtt{w}_* = u_*$, $\widehat{\mathtt{w}}_\lambda = \widehat{u}$, $n = m$, $\mathcal{H} = \mathbb{R}^{d+1}$, $\mathcal{B} = \mathcal{Z}$, and $\lambda = \mu$. □

We next restate Lemma 2 from Appendix B of Della Vecchia et al. (2021).

Lemma 6.3 (Lemma 2, Della Vecchia et al. (2021))**.** *Suppose the assumptions (A1), (A2), and (A3) hold. Then for sample size n and all $\delta \in (0, 1)$, with probability at least $1 - \delta$, the following ineqaulity holds.*

$$\mathfrak{L}(\mathtt{w}) - \hat{\mathfrak{L}}(\mathtt{w}) \leq \frac{4G(1 + \alpha\|\mathtt{w}\|)}{\sqrt{n}}\Big(1 + \sqrt{\ln(2 + \log_2(1 + \alpha\|\mathtt{w}\|)+) + \ln(1/\delta)}\Big).$$

We use this lemma to quantify the gap between the empirical and population risk for any hypothesis w when side information is imputed by a classifier $\phi : \mathcal{X} \to \mathcal{Z}$.

Lemma 6.4. *Let $w := (w_c, w_x, w_z) \in \mathbb{R}^3$. Let $\hat{\phi}$ be defined in (8) and let κ be defined in (1). Let $D = \{(x_i, y_i)\}_{i=1}^n$ and $\delta \in (0,1)$. Then, with probability at least $1-\delta$, we have*

$$R^{\hat{\phi}}(w) - \hat{R}^{\hat{\phi}}(w) \leq \frac{4(1+\sqrt{\kappa^2+2}\|w\|)}{\sqrt{n}}\Big(1 + \sqrt{\ln(2+\log_2(1+\sqrt{\kappa^2+2}\|w\|)) + \ln(1/\delta)}\Big).$$

Proof. Identify $\mathfrak{L}$ with $R^{\hat{\phi}}$, $\hat{\mathfrak{L}}$ with $\hat{R}^{\hat{\phi}}$, $\mathcal{A}$ with $\mathbb{R}^{d+2}$, $\mathfrak{B}$ with $\mathcal{Y}$, B with Y, $\mathcal{A}$ with $\{1\} \times \mathcal{X} \times \mathcal{Z}$, A with $(1, X, \hat{\phi}(X))$, $\mathtt{w}$ with w, and α with $\sqrt{\kappa^2+2}$. Then apply Lemma 2 in Appendix B of Della Vecchia et al. (2021). □

Lemma 6.5. *Let (X, Y) be a random variable over $\mathbb{R}^k \times \{-1, 1\}$ such that it is linearly non-separable, i.e., for all $a \in \mathbb{R}^{k+1}$, $\mathbb{P}(\mathsf{Y}(a_c + a_x^\top \mathsf{X}) \leq 0) > 0$. Let the logistic population risk be defined as*

$$L(a) := \mathbb{E}[-\ln(\sigma(\mathsf{Y}(a_c + a_x^\top \mathsf{X})))].$$

Then the optimal weight $a^\star := \arg\min_a L(a)$ satisfies $\|a^\star\| < \infty$.

Proof. We first show that $L(a) \to \infty$ as $\|a\| \to \infty$. This, together with the fact that $L(a^\star) \leq L(0) = \ln(2)$, implies that the minimizer must lie in a bounded set, proving $\|a^\star\| < \infty$. Let a be an arbitrary direction such that $\|a\| = 1$, and consider the sequence $a_k := ka$ where $k \in \mathbb{N}$. Then $\|a_k\| \to \infty$ as $k \to \infty$. From the non-separability assumption we have $\mathbb{P}(\mathsf{Y}(a_c + a_x^\top \mathsf{X}) \leq 0) = \epsilon > 0$. Note that if $\mathsf{Y}(u_c + u_x^\top \mathsf{X}) < 0$, then, $k\mathsf{Y}(u_c + u_x^\top \mathsf{X}) < 0$ for $k \in \mathbb{N}$, thus $\mathbb{P}(k\mathsf{Y}(a_c + a_x^\top \mathsf{X}) \leq 0) = \epsilon > 0$. We lower bound the loss $L(a_k)$.

$$L(a_k) = \mathbb{E}[-\ln(\sigma(k\mathsf{Y}(a_c + a_x^\top \mathsf{X})))] \geq -\ln(\sigma(k\mathsf{Y}(a_c + a_x^\top \mathsf{X})) \cdot \epsilon, \text{ where } \mathsf{Y}(a_c + a_x^\top \mathsf{X}) \leq 0.$$

We use the inequality $-\ln(\sigma(z)) = \ln(1 + e^{-z}) \geq -z$, which holds for all $z \in \mathbb{R}$. Applying this we get:

$$L(a_k) \geq -\ln(\sigma(k\mathsf{Y}(a_c + a_x^\top \mathsf{X})) \cdot \epsilon \geq -k\mathsf{Y}(a_c + a_x^\top X) \cdot \epsilon \text{ where } \mathsf{Y}(a_c + a_x^\top \mathsf{X}) \leq 0.$$

Thus as $k \to \infty$, $L(a_k) \to \infty$. Since a was an arbitrary unit vector, we have $L(a) \to \infty$ as $\|a\| \to \infty$. □

7 Appendix II: Bounding optimal weights

Algorithm 1 trains on the partitioned datasets D and S, and returns a classifier with weights $\hat{w}_\lambda \in \mathbb{R}^{d+2}$. In the forthcoming proof of Theorem 4.1, we will show that the excess risk associated with $\hat{w}_\lambda$ admits an upper bounded involving terms of order $\frac{\|\hat{w}_\lambda\|}{\sqrt{n}}$. However, since $\hat{w}_\lambda$ is a random variable, we do not know how terms like $\frac{\|\hat{w}_\lambda\|}{\sqrt{n}}$ scale with sample size n. Thus we seek to show that, with high probability, $\|\hat{w}_\lambda\|$ lies in a bounded subset of $\mathbb{R}^{d+2}$. A crucial property required for this analysis is the non-separability of the distribution (see (1)). The lemmas in this section will conclude by providing an upper bound on $\|\hat{w}_\lambda\|$ and will also establish $\mathsf{W} < \infty$ (defined in (16)) as promised.

We now give the definition of strong convexity.

Definition 7.1. A differentiable function $f : \mathfrak{Dom} \subseteq \mathbb{R}^k \to \mathbb{R}$ is strongly convex if there exists a strong convexity parameter $\alpha > 0$ such that

$$f(w) \geq f(w') + \nabla f(w)^\top (w - w') + \alpha\|w - w'\|^2$$

for all $w, w' \in \mathfrak{Dom}$.

An equivalent definition in terms of Hessian is given as follows.

Definition 7.2. A twice differentiable function $f : \mathfrak{Dom} \subseteq \mathbb{R}^k \to \mathbb{R}$ is strongly convex if there exists a strong convexity parameter $\mu > 0$ such that

$$z^\top (\nabla^2 f(w)) z \geq \mu\|z\|^2,$$

for all $z, w \in \mathfrak{Dom}$.

To show that $\|\hat{w}_\lambda\|$ is upper-bounded with high probability, we establish that $\hat{\phi}(x)$ is close to $\phi^\star(x)$ on average over the distribution of X. For this, we prove an intermediate result that shows that logistic population risk $L(u)$ is strongly convex in the sublevel set

$$\mathfrak{Dom} := \Big\{u : L(u) - L(u^\star) \leq \epsilon_{\mathfrak{u}}\Big\}, \tag{23}$$

where $\epsilon_{\mathfrak{u}}$ is given in (22). Note that $u^\star \in \mathfrak{Dom}$ by definition. We first show that $\mathfrak{Dom}$ is a bounded subset of $\mathbb{R}^{d+1}$ and $\hat{u} \in \mathfrak{Dom}$ with high probability in the following lemma.

Lemma 7.1. *Let S be given by* (2)*, $\mathfrak{Dom}$ given in* (23)*, and $\hat{u}$ be defined in* (7)*. Let $\delta \in (0,1)$. Then there exists $C > 0$, such that $\mathfrak{Dom} \subseteq \{u : \|u\| \leq C\}$, and with probability at least $1-\delta$, we have $\hat{u} \in \mathfrak{Dom}$.*

Proof. Let $u^\star$ be given by (5). From the non-separability assumption on (X, Z) given in (1), $\|u^\star\| < \infty$. From Lemma 6.2, we have with probability at least $1-\delta$,

$$L(\hat{u}) - L(u^\star) \leq \epsilon_{\mathfrak{u}}. \tag{24}$$

From the non-separability assumption on (X, Z), we have $L(u) \to \infty$ as $\|u\| \to \infty$. Therefore $\mathfrak{Dom}$ is bounded. Hence there exists $C > 0$ such that $\mathfrak{Dom} \subseteq \{u : \|u\| \leq C\}$. This implies that with probability at least $1-\delta$, $\hat{u} \in \mathfrak{Dom}$. □

Next, we show that $L(u)$ is strongly convex on $\mathfrak{Dom}$ in the following lemma.

Lemma 7.2. *Let $\mathfrak{Dom}$ be given in* (23)*. Then the logistic risk $L(u)$ is strongly convex on $\mathfrak{Dom}$*

Proof. The Hessian of the logistic risk $L(u)$ is

$$\nabla^2 L(u) = \mathbb{E}_{(X,Z)}\left[\sigma(Zs)\big(1-\sigma(Zs)\big)(1,X)(1,X)^\top\right],$$

where $s = u_c + u_x^\top X$. From (1), $\mathbb{E}[(1,X)(1,X)^\top] \succ 0$ (positive definite), and $\|(1,X)\|$ is bounded. Furthermore, by assumption, for all $u \in \mathfrak{Dom}$, $\|u\|$ is bounded. Therefore $0 < \sigma(Zs)(1-\sigma(Zs)) < \infty$. Thus, the Hessian is positive definite for all $u \in \mathfrak{Dom}$, and $L(u)$ is strongly convex. □

Using this, we now show in the following lemma that $\hat{\phi}(x)$ is close to $\phi^\star(x)$ on average over the distribution of X. Specifically, we quantify the expected disagreement between the learned classifier $\hat{\phi} : \mathcal{X} \to \mathcal{Z}$ and the optimal classifier $\phi^\star : \mathcal{X} \to \mathcal{Z}$ under the joint distribution of (X, Z).

Lemma 7.3. *Let $\hat{\phi}$ and $\phi^\star$ be defined as in equation* (8) *and let S be defined in* (2)*. Let $\delta \in (0,1)$. Then, with probability at least $1-\delta$, we have*

$$\mathbb{E}_X\Big[|\hat{\phi}(X) - \phi^\star(X)|\Big] \leq \mathcal{O}\Big(\Big(\frac{1}{\sqrt{\|u^\star\|^3}}\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big).$$

Proof. From (8), $u^\star$ and $\hat{u}$ are the weights corresponding to $\phi^\star(x)$ and $\hat{\phi}(x)$. We first upper bound $\|\hat{u} - u^\star\|$ and then use the Lipschitz property to upper bound $\mathbb{E}_{X,Z}[|\hat{\phi} - \phi^\star|]$.

Claim. With probability at least $1-\delta$, we have

$$\|\hat{u} - u^\star\| \leq \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big).$$

Proof of the claim. From Lemma 7.2, we have $L(u)$ is strongly convex on $\mathfrak{Dom}$ given in (23). From Let α be the strong convexity parameter of L. Then from Definition 7.1, we have with probability at least $1-\delta$,

$$L(\hat{u}) \geq L(u^\star) + \alpha\|\hat{u} - u^\star\|^2. \tag{25}$$

From Lemma 6.2, we have with probability at least $1-\delta$,

$$L(\hat{u}) - L(u^\star) \leq \epsilon_{\mathrm{u}} = \mathcal{O}\Big(\|u^\star\|\sqrt{\frac{\ln(1/\delta)}{m}}\Big). \tag{26}$$

Substituting (26) into (25), we get

$$\alpha\|\hat{u} - u^\star\|^2 \leq L(\hat{u}) - L(u^\star) \leq \mathcal{O}\Big(\|u^\star\|\sqrt{\frac{\ln(1/\delta)}{m}}\Big).$$

Thus, with probability at least $1-\delta$,

$$\|\hat{u} - u^\star\| \leq \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big).$$

□

We now upper bound $\mathbb{E}_X\Big[|\hat{\phi}(X) - \phi^\star(X)|\Big]$. Let $\hat{s}(x) := \hat{u}_c + \hat{u}_x^\top x$ and $s^\star(x) := u_c^\star + {u_x^\star}^\top x$. Using the Lipschitz property of the sigmoid function, where both x and u's are treated as variables, with Lipschitz constant K, we have

$$|\sigma(\hat{s}(x)) - \sigma(s^\star(x))| \leq K\sqrt{1+\kappa^2}\|\hat{u} - u^\star\| \leq \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big),$$

where we have used the fact that $\|(1,x)\| \leq \sqrt{1+\kappa^2}$ from the assumptions that the random variable of X is bounded almost surely (see (1)). This implies that if either

$$s^\star(x) > \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big) \text{ or } s^\star(x) < -\mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big),$$

holds, then $\phi^\star(x) = \hat{\phi}(x)$. We now show that the event $\mathcal{E} := \Big\{|s^\star(X)| \leq \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big)\Big\}$ has small probability. The set $\mathcal{E}$ can be rewritten as

$$\mathcal{E} = \Big\{x : -\mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big) =: -\epsilon \leq u_c^\star + {u_x^\star}^\top x \leq \epsilon := \mathcal{O}\Big(\sqrt{\|u^\star\|}\Big(\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big)\Big\}.$$

This set lies between two hyperplanes. We can decompose $x = x_\perp + x_\parallel$, where $x_\parallel = \beta u_x^\star$ and $\langle x_\perp, u_x^\star\rangle = 0$. Thus, we get the simplification

$$\mathcal{E} = \Big\{x : x = x_\perp + x_\parallel, \langle x_\perp, u_x^\star\rangle = 0, \frac{-\epsilon - u_c^\star}{\|u_x^\star\|^2} \leq \beta \leq \frac{\epsilon - u_c^\star}{\|u_x^\star\|^2}\Big\},$$

Let $f(x) \equiv f(x_\perp, x_\parallel)$ be the probability density function of the random variable X. Then $\mathbb{P}_X(\mathcal{E})$ is given by

$$\mathbb{P}_X(\mathcal{E}) = \int_{(-\epsilon-u_c^\star)/\|u_x^\star\|^2}^{(\epsilon-u_c^\star)/\|u_x^\star\|^2}\int_{-\infty}^{\infty} f(x_\perp, x_\parallel)dx_\perp dx_\parallel = \int_{(-\epsilon-u_c^\star)/\|u_x^\star\|^2}^{(\epsilon-u_c^\star)/\|u_x^\star\|^2} g(x_\parallel)dx_\parallel,$$

where $g(x_\parallel) = \int_{-\infty}^{\infty} f(x_\perp, x_\parallel)dx_\parallel$. Using the property of the probability density function from (1), we have $\int_{-\epsilon}^{\epsilon} g(x_\parallel)dx_\parallel \leq K\frac{\epsilon}{\|u_x^\star\|^2}$, for some $K > 0$. Therefore, with probability at least $1-2\delta$, we have

$$\mathbb{P}_X\Big(\hat{\phi}(X) \neq \phi^\star(X)\Big) \leq \mathbb{P}_X(\mathcal{E}) \leq \mathcal{O}\Big(\Big(\frac{1}{\sqrt{\|u^\star\|^3}}\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big),$$

which implies

$$\mathbb{E}_X\Big[|\hat{\phi}(X) - \phi^\star(X)|\Big] \leq \mathcal{O}\Big(\Big(\frac{1}{\sqrt{\|u^\star\|^3}}\frac{\ln(1/\delta)}{m}\Big)^{\frac{1}{4}}\Big).$$

This completes the proof. □

Finally, using the previous result, we now upper bound $\|\hat{w}_\lambda\|$ in the following lemma.

Lemma 7.4. *Let $\|\hat{w}_\lambda\|$ be defined in (15) and let $\delta \in (0,1)$. Then with probability at least $1-2\delta$, their exists a bounded subset of $\mathbb{R}^{d+2}$ such that $\|\hat{w}_\lambda\|$ lies in that subset for all $\lambda \in R_+^3$, i.e.,*

$$\sup_{(\lambda_c,\lambda_x\lambda_z)} \|\hat{w}_\lambda\| < \infty \text{ with probability } 1-2\delta.$$

Proof. We first show that for fixed $w \in \mathbb{R}^{d+2}$, the values of $R^{\phi^\star}(w)$ and $R^{\hat{\phi}}(w)$ are close.

Claim. Let $\phi^\star$ and $\hat{\phi}$ be defined in (8). Then with probability at least $1-\delta$,

$$|R^{\hat{\phi}}(w) - R^{\phi^\star}(w)| \leq \mathcal{O}\Big(\sqrt{\frac{\ln(1/\delta)}{m}}\Big).$$

Proof of the Claim. From the Lipschitz property of $\ell(\cdot,\cdot,Y;w)$ with respect to its second argument (fixing Y), let $K_Y > 0$ denote the Lipschitz constant, which may depend on Y, and define $K := \max_Y K_Y$. Then,

$$\begin{aligned}
|R^{\hat{\phi}}(w) - R^{\phi^\star}(w)| &= \Big|\mathbb{E}_{(X,Y)}\big[\ell(X,\hat{\phi}(X),Y;w) - \ell(X,\phi^\star(X),Y;w)\big]\Big| \\
&\leq \mathbb{E}_{(X,Y)}\big[|\ell(X,\hat{\phi}(X),Y;w) - \ell(X,\phi^\star(X),Y;w)|\big] \\
&\leq \mathbb{E}_Y\Big[\mathbb{E}_{X|Y}\big[K|\hat{\phi}(X) - \phi^\star(X)|\big]\Big] \leq \mathcal{O}\Big(\sqrt{\frac{\ln(1/\delta)}{m}}\Big),
\end{aligned}$$

where the first inequality follows from Jensen's lemma, and the second inequality uses the Lipschitz property of $\ell(\cdot,\cdot,Y;w)$ and the last inequality follows from Lemma 7.3. □

Let D and S be as defined in (2) and let

$$H_1(D,S) := \{w : \hat{R}^{\hat{\phi}}(w) \leq \ln(2)\}.$$

From Lemma 6.4, we have with probability at least $1-\delta$,

$$|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq C\|w\|\sqrt{\frac{\ln(\ln(\|w\|) + \ln(1/\delta))}{n}},$$

for some $C > 0$. Let

$$H_2(S) := \Big\{w : R^{\hat{\phi}}(w) + C\|w\|\sqrt{\frac{\ln(\ln(\|w\|) + \ln(1/\delta))}{n}} \leq \ln(2)\Big\}.$$

Thus with probability at least $1-\delta$ (keeping S fixed), we have $H_1(S,D) \subseteq H_2(S)$. Let

$$H := \Big\{w : R^{\phi^\star}(w) + C\|w\|\sqrt{\frac{\ln(\ln(\|w\|) + \ln(1/\delta))}{n}} \leq \ln(2) + C'\sqrt{\frac{\ln(1/\delta)}{m}}\Big\}, \text{ where } C' > 0.$$

Using the Claim, with probability at least $1-\delta$, we have $H_2(S) \subseteq H$ and thus with probability at least at least $1-2\delta$, we have $H_1(D,S) \subseteq H$. Note that with probability at least $1-\delta$, $\hat{w}_\lambda \in H$ for all λ, as $\hat{R}^{\hat{\phi}}(\hat{w}_\lambda) \leq \hat{R}^{\hat{\phi}}(0) = \ln(2)$. Since $R^{\phi^\star}(w) + C\|w\|\sqrt{\frac{\ln(\ln(\|w\|)+\ln(1/\delta))}{n}}$ is a continuous in w, and H is a preimage of a closed set under a continuous map, it follows that H is closed. Furthermore, H is bounded because from the non-separability condition on (X,Y,Z) given in (1), $R^{\phi^\star}(w)$ is coercive, i.e., $R^{\phi^\star}(w) \to \infty$ as $\|w\| \to \infty$. Thus $\sup H$ exists and therefore with probability at least $1-2\delta$, $\|\hat{w}_\lambda\|$ is uniformly bounded. □

To show $\mathsf{W} < \infty$ (defined in (16)), we also need to show that with high probability, $\|w^\star_{\phi^\star,\lambda}\|$ and $\|w^\star_\lambda\|$ lies in a bounded subset of $\mathbb{R}^{d+2}$. We start with $\|w^\star_{\phi^\star,\lambda}\|$. The next lemma formalizes this.

Lemma 7.5. *Let $\phi^\star$ be defined in* (8) *and let $W_{\phi^\star} : (\mathbb{R}_{\geq 0} \cup \{\infty\})^3 \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ be the function defined by*

$$W_{\phi^\star}(\lambda_c, \lambda_x, \lambda_z) := \|w^\star_{\phi^\star,\lambda}\|, \tag{27}$$

where $w^\star_{\phi^\star,\lambda}$ is defined in (14) *(with $\phi = \phi^\star$). Then for all $(\lambda_c, \lambda_x, \lambda_z) \in \mathbb{R}^3_+$,*

$$W_{\phi^\star} := \sup_{(\lambda_c,\lambda_x,\lambda_z)} W_{\phi^\star}(\lambda_c, \lambda_x, \lambda_z) < \infty.$$

Proof. Let $\lambda = (\lambda_c, \lambda_x, \lambda_z)$. From the non-separability assumption on (X, Y, Z) given in (1), the population risk $R^{\phi^\star}(w)$ is coercive, i.e.,

$$R^{\phi^\star}(w) \to \infty \quad \text{as } \|w\| \to \infty.$$

Therefore, for every λ, the regularized risk

$$R^{\phi^\star}_\lambda(w) := R^{\phi^\star}(w) + \lambda_c|w_c|^2 + \lambda_x|w_x|^2 + \lambda_z|w_z|^2$$

is also coercive, which ensures that the minimum is attained at $w^\star_{\phi^\star,\lambda}$ with $\|w^\star_{\phi^\star,\lambda}\| < \infty$. Moreover, since $w^\star_{\phi^\star,\lambda}$ minimizes $R^{\phi^\star}_\lambda$, we have

$$R^{\phi^\star}_\lambda(w^\star_{\phi^\star,\lambda}) \leq R^{\phi^\star}_\lambda(0) = \ln 2 \text{ which implies } R^{\phi^\star}(w^\star_{\phi^\star,\lambda}) \leq \ln 2,$$

Define the set

$$H := \{w \in \mathbb{R}^3 : R^{\phi^\star}(w) \leq \ln 2\}.$$

Then for all λ, we have $w^\star_{\phi^\star,\lambda} \in H$. We now show that H is compact. Since $R^{\phi^\star}(w)$ is continuous in w, and H is the preimage of a closed set under a continuous map, it follows that H is closed. To show boundedness, suppose for contradiction that H is unbounded. Then there exists a direction w such that $tw \in H$ for all $t > 0$. But coercivity of $R^{\phi^\star}$ implies that $R^{\phi^\star}(tw) \to \infty$ as $t \to \infty$, contradicting the assumption that $tw \in H$. Hence, H must be bounded. Since $w^\star_{\phi^\star,\lambda} \in H$ for all λ, and H is compact, we conclude

$$\sup_\lambda \|w^\star_{\phi^\star,\lambda}\| \leq \sup_{w \in H} \|w\| < \infty,$$

which completes the proof.

□

Arguing similarly as in Lemma 7.5, we can also show that with high probability, $\|w^\star_\lambda\|$ lies in a bounded subset of $\mathbb{R}^{d+2}$, as given in the next lemma.

Lemma 7.6. *Let $W : (\mathbb{R}_{\geq 0} \cup \{\infty\})^3 \to \mathbb{R}_{\geq 0} \cup \{\infty\}$ be the function defined by*

$$W(\lambda_c, \lambda_x, \lambda_z) := \|w^\star_\lambda\|, \tag{28}$$

where $w^\star_\lambda$ is defined in (12)*. Then for all $(\lambda_c, \lambda_x, \lambda_z) \in \mathbb{R}^3_+$,*

$$W = \sup_{(\lambda_c,\lambda_x,\lambda_z)} W(\lambda_c, \lambda_x, \lambda_z) < \infty.$$

Thus combining Lemma 7.5, Lemma 7.6, Lemma 7.1, and Lemma 7.4, we get $\mathsf{W} < \infty$ with high probability.

Proposition 1. *Let $\mathsf{W} \in \mathbb{R}$ be defined as*

$$\mathsf{W} := \max\{W, W_{\phi^\star}, \sup_\lambda \|\hat{w}_\lambda\|, \|w^\star\|, \|u^\star\|, \|\hat{u}\|\}.$$

and let $\delta \in (0, 1)$. Then with probability at least $1 - 2\delta$, $\mathsf{W} < \infty$.

8 Appendix III: Other Useful Lemmas

This section contains the remaining lemmas required for proving the main theorem (Theorem 4.1). The following lemma calculates the gap in risk when Z is substituted with $\phi : \mathcal{X} \to \mathcal{Z}$.

Lemma 8.1. *Let $w := (w_c, w_x, w_z) \in \mathbb{R}^{d+2}$ and let $\hat{\phi}$ be defined in* (8)*. Then:*

$$|R^{\hat{\phi}}(w) - R(w)| \leq 2|w_z| L_{(0,1)}(\hat{u}).$$

Proof. For a fixed (x, y) and w, the logistic loss function $\ell(x, t, y; w) := \ln(1 + \exp(-y(w_c + w_x^\top x + w_z t)))$ is Lipschitz in t, with a Lipschitz bound given by

$$K \leq \sup_{x,t,y} (1 - \sigma(y(w_c + w_x^\top x + w_z t)))(-y w_z) \leq |w_z|.$$

Thus, for $t_1 = z$ and $t_2 = \hat{\phi}(x)$, we have:

$$|\ell(x, z, y; w) - \ell(x, \hat{\phi}(x), y; w)| \leq |w_z||z - \hat{\phi}(x)|.$$

From direct calculation, it follows that $\mathbb{E}_{(X,Z)}|z - \hat{\phi}(x)| = 2L_{(0,1)}(\hat{u})$, where $L_{(0,1)}(\cdot)$ denotes the zero-one risk.

We now bound $|R^{\hat{\phi}}(w) - R(w)|$. Expanding $R^{\hat{\phi}}(w)$ and $R(w)$ in terms of the logistic loss function, we get:

$$|R^{\hat{\phi}}(w) - R(w)| = \Big|\mathbb{E}_{(X,Y,Z)}[\ell(x, y, z, w)] - \mathbb{E}_{(X,Y,Z)}[\ell(x, y, \hat{\phi}(x), w)]\Big|.$$

Using Jensen's inequality, we have

$$\mathbb{E}_{(X,Y,Z)}\Big[|\ell(X, Y, \hat{\phi}(X), w) - \ell(X, Y, Z, w)|\Big] \geq \Big|\mathbb{E}_{(X,Y,Z)}[\ell(x, z, y, w)] - \mathbb{E}_{(X,Y,Z)}[\ell(x, \hat{\phi}(x), y, w)]\Big|.$$

Note that

$$\mathbb{E}_{(X,Y)}[\ell(x, \hat{\phi}(x), y, w)] = \mathbb{E}_{(X,Y,Z)}[\ell(x, \hat{\phi}(x), y, w)], \text{ and } \mathbb{E}_{(X,Z)}\Big[|Z - \hat{\phi}(X)|\Big] = \mathbb{E}_{(X,Y,Z)}\Big[|Z - \hat{\phi}(X)|\Big].$$

Thus, we get

$$|R^{\hat{\phi}}(w) - R(w)| \leq \mathbb{E}_{(X,Y,Z)}\Big[|\ell(X, Y, \hat{\phi}(X), w) - \ell(X, Y, Z, w)|\Big] \leq 2|w_z| L_{(0,1)}(\hat{u}).$$

□

The next lemma establishes the effect of the regularizer on w_z.

Lemma 8.2. *Let $\lambda := (\lambda_c, \lambda_x, \lambda_z) \in \mathbb{R}^3$ and let $\hat{w}_\lambda$ and $w_\lambda^\star$ be defined in* (15) *and* (12)*, respectively. Then:*

$$|\hat{w}_{\lambda,z}| \leq \sqrt{\frac{\ln(2)}{\lambda_z}} \quad \textit{and} \quad |w_{\lambda,z}^\star| \leq \sqrt{\frac{\ln(2)}{\lambda_z}}.$$

Proof. From the definition of $\hat{w}_\lambda$, we have:

$$\lambda_z |\hat{w}_{\lambda,z}|^2 \leq \hat{R}_\lambda^{\hat{\phi}}(\hat{w}_\lambda) \leq \hat{R}_\lambda^{\hat{\phi}}(0) = \ln(2),$$

which implies:

$$|\hat{w}_{\lambda,z}| \leq \sqrt{\frac{\ln(2)}{\lambda_z}}.$$

Similarly, from the definition of $w_\lambda^\star$, we have:

$$\lambda_z |w_{\lambda,z}^\star|^2 \leq R_\lambda(w_\lambda^\star) \leq R_\lambda(0) = \ln(2),$$

which implies:

$$|w^\star_{\lambda,z}| \leq \sqrt{\frac{\ln(2)}{\lambda_z}}.$$

□

The following lemma is an application of McDiarmid's inequality to $\hat{R}^{\hat{\phi}}(w)$ and $R^{\hat{\phi}}(w)$.

Lemma 8.3. *Let $\hat{\phi}$ be defined in (8) and let $w := (w_c, w_x, w_z) \in \mathbb{R}^3$. Let D be defined in (2) and let κ be defined in (1). Let, for a fixed y, K_y be the Lipschitz constant of $\ell(\cdot, y, \cdot; \cdot)$ and let $K := \max_y K_y$. Let $\delta \in (0, 1)$. Then, with probability at least $1 - \delta$, we have:*

$$|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq (\ln(2) + K\sqrt{\kappa^2 + 2}\|w\|)\sqrt{\frac{\ln(2/\delta)}{n}}. \tag{29}$$

Proof. For a fixed $w \in \mathbb{R}^{d+2}$, let $Q_i := (X_i, Y_i)$ and $f(Q_1, \cdots, Q_n) := \hat{R}^{\hat{\phi}}(w)$. We will apply McDiarmid's inequality to $f(\cdots, Q_i, \cdots)$ to get the upper bound in the lemma. To apply McDiarmid's inequality, we must show that $|f(\cdots, Q_i, \cdots) - f(\cdots, Q'_i, \cdots)|$ is bounded for each $1 \leq i \leq n$.

We have

$$f(Q_1, \cdots, Q_i, \cdots, Q_n) - f(Q_1, \cdots, Q'_i, \cdots, Q_n) = \ell(x_i, y_i, \hat{\phi}(x_i); w) - \ell(x_{i'}, y_{i'}, \hat{\phi}(x_{i'}); w).$$

Furthermore

$$|\ell(x_i, y_i; \hat{\phi}(x_i); w)| \leq |\ell(x_i, y_i; \hat{\phi}(x_i); 0)| + |\ell(x_i, y_i; \hat{\phi}(x_i); w) - \ell(x_i, y_i; \hat{\phi}(x_i); 0)|.$$

Note that $\ell(x, \hat{\phi}(x), y; w)$ can be rewritten as $\ell(y, s_w)$ where $s_w : w_c + w_x x + w_z \hat{\phi}(x)$. Using the Lipschitz property of ℓ and the definition of K given in the statement of the theorem, we have

$$|\ell(y, s_w) - \ell(y, s_{w'})| \leq K\|s_w - s_{w'}\|.$$

Substituting this, we get

$$|\ell(x_i, y_i; \hat{\phi}(x_i); w)| \leq \ln(2) + K|\langle(1, x_i, \hat{\phi}(x_i)), w\rangle| \leq \ln(2) + K\sqrt{\kappa^2 + 2}\|w\|.$$

Thus $|f(\cdots, Q_i, \cdots) - f(\cdots, Q'_i, \cdots)| \leq \ln(2) + \sqrt{\kappa^2 + 2}\|w\|$, and therefore, we can now apply McDiarmid's inequality to $f(Q_1, \cdots, Q_n)$. Hence, with probability at least $1 - \delta$,

$$|\hat{R}^{\hat{\phi}}(w) - R^{\hat{\phi}}(w)| \leq (\ln(2) + K\sqrt{\kappa^2 + 2}\|w\|)\sqrt{\frac{\ln(2/\delta)}{n}}.$$

□

The next lemma shows that the third component of a regularizer is inversely proportional to the corresponding optimal weight component. Furthermore, it also shows that their corresponding risks are inversely proportional.

Lemma 8.4. *Let $\lambda = (\lambda_c, \lambda_x, \lambda_z)$ and $\theta = (\lambda_c, \lambda_x, \theta_z)$ with $\lambda_z \leq \theta_z$. Then, by keeping λ_x and λ_c fixed, we have:*

$$|w^\star_{\theta,z}| \leq |w^\star_{\lambda,z}|. \tag{30}$$

If $\lambda_c = \lambda_x = \frac{1}{\sqrt{n}}$, and keeping λ_z, θ_z fixed, then

$$R(w^\star_\lambda) \leq R(w^\star_\theta) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big). \tag{31}$$

Furthermore, we also have:

$$|w^\star_{\phi^\star,\theta,z}| \leq |w^\star_{\phi^\star,\lambda,z}|, \tag{32}$$

and

$$R^{\phi^\star}(w^\star_\lambda) \leq R^{\phi^\star}(w^\star_\theta) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big), \tag{33}$$

where W is defined in (16).

Proof. Let $\lambda = (\lambda_c, \lambda_x, \lambda_z)$ and $\theta = (\lambda_c, \lambda_x, \theta_z)$. If $\lambda_z < \theta_z$, then we have:

$$R(w^\star_\lambda) + \lambda_c |w^\star_{\lambda,c}|^2 + \lambda_x \|w^\star_{\lambda,x}\|^2 + \lambda_z |w^\star_{\lambda,z}|^2 \leq R(w^\star_\theta) + \lambda_c |w^\star_{\theta,c}|^2 + \lambda_x \|w^\star_{\theta,x}\|^2 + \lambda_z |w^\star_{\theta,z}|^2,$$

and:

$$R(w^\star_\theta) + \lambda_c |w^\star_{\theta,c}|^2 + \lambda_x \|w^\star_{\theta,x}\|^2 + \theta_z |w^\star_{\theta,z}|^2 \leq R(w^\star_\lambda) + \lambda_c |w^\star_{\lambda,c}|^2 + \lambda_x \|w^\star_{\lambda,x}\|^2 + \theta_z |w^\star_{\lambda,z}|^2.$$

This implies

$$(\lambda_z - \theta_z)|w^\star_{\lambda,z}|^2 \leq (\lambda_z - \theta_z)|w^\star_{\theta,z}|^2,$$

or equivalently

$$|w^\star_{\theta,z}| \leq |w^\star_{\lambda,z}| \tag{34}$$

proving (30).

Let $\lambda = (\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \lambda_z)$ and $\nu = (0, 0, \lambda_z)$ and λ_z be fixed. Then we have

$$R(w^\star_\nu) + \lambda_z |w^\star_{\nu,z}|^2 \leq R(w^\star_\lambda) + \lambda_z |w^\star_{\lambda,z}|^2$$

and

$$R(w^\star_\lambda) + \frac{1}{\sqrt{n}}(|w^\star_{\lambda,c}|^2 + \|w^\star_{\lambda,x}\|^2) + \lambda_z |w^\star_{\lambda,z}|^2 \leq R(w^\star_\nu) + \frac{1}{\sqrt{n}}(|w^\star_{\nu,c}|^2 + \|w^\star_{\nu,x}\|^2) + \lambda_z |w^\star_{\nu,z}|^2.$$

This implies

$$\frac{1}{\sqrt{n}}(|w^\star_{\lambda,c}|^2 + \|w^\star_{\lambda,x}\|^2) \leq \frac{1}{\sqrt{n}}(|w^\star_{\nu,c}|^2 + \|w^\star_{\nu,x}\|^2) = \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big), \tag{35}$$

where W is defined in (16). Now consider the expression for $R(w^\star_\lambda)$ and $R(w^\star_\theta)$.

$$R(w^\star_\lambda) + \frac{1}{\sqrt{n}}(|w^\star_{\lambda,c}|^2 + \|w^\star_{\lambda,x}\|^2) + \lambda_z |w^\star_{\lambda,z}|^2 \leq R(w^\star_\theta) + \frac{1}{\sqrt{n}}(|w^\star_{\theta,c}|^2 + \|w^\star_{\theta,x}\|) + \lambda_z |w^\star_{\theta,z}|^2.$$

Using (34), we get

$$R(w^\star_\lambda) + \frac{1}{\sqrt{n}}(|w^\star_{\lambda,c}|^2 + \|w^\star_{\lambda,x}\|^2) \leq R(w^\star_\theta) + \frac{1}{\sqrt{n}}(|w^\star_{\theta,c}|^2 + \|w^\star_{\theta,x}\|^2)^2.$$

But applying (35), we get

$$R(w^\star_\lambda) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big) \leq R(w^\star_\theta) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big) \text{ or } R(w^\star_\lambda) \quad \leq R(w^\star_\theta) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big). \tag{36}$$

A similar argument can be used to prove

$$|w^\star_{\phi^\star,\theta,z}| \leq |w^\star_{\phi^\star,\lambda,z}|, \tag{37}$$

and

$$R^{\phi^\star}(w^\star_\lambda) \leq R^{\phi^\star}(w^\star_\theta) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

as claimed. □

Corollary 8.4.1. *Let* $\lambda = \Big(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \lambda_z\Big)$ *for some* $n > 0$. *Then*

$$R(w^\star_\lambda) \leq T(v^\star) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

Proof. Let $\nu = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \infty\right)$ and

$$v_\nu^\star = (v_{\nu,c}^\star, v_{\nu,x}^\star, v_{\nu,z}^\star) = (v_{\nu,c}^\star, v_{\nu,x}^\star, 0) := \arg\min_v T(v) + \nu_c|v_c|^2 + \nu_x\|v_x\|^2 + \nu_z|v_z|^2.$$

Note that $v^\star$ can be written as $v^\star = (v_c^\star, v_x^\star, 0)$ and thus $v^\star$ can be thought of as a solution of a regularisation problem with regularizer $(0, 0, \infty)$. Thus, using Lemma 8.4 equation (35), we have

$$R(v_\nu^\star) \leq T(v^\star) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

From Lemma 8.4 equation (31), we have

$$R(w_\lambda^\star) \leq R(v_\nu^\star) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

Thus, combining these two equations, we have

$$R(w_\lambda^\star) \leq T(v^\star) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

□

9 APPENDIX IV: PROOF OF THE MAIN THEOREM

We now present the proof of Theorem 4.1.

Proof of Theorem 4.1. Let $\delta \in (0, 1)$. The LHS of (17) can be decomposed as:

$$\begin{aligned} R(\hat{w}_\lambda) - R(w^*) = &\underbrace{R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda)}_{(A)} + \underbrace{R^\phi(\hat{w}_\lambda) - R^\phi(w_\lambda^\star)}_{(B)} \\ &+ \underbrace{R^\phi(w_\lambda^\star) - R(w_\lambda^\star)}_{(C)} + \underbrace{R(w_\lambda^\star) - R(w^*)}_{(D)}. \end{aligned}$$

We will first address the sum of terms (A) and (C).

Claim 1. With probability at least $1 - 2\delta$,

$$R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda) + R^\phi(w_\lambda^\star) - R(w_\lambda^\star) \leq \ln(4)\left(\frac{C_2}{\sqrt{m}} + L(u^\star)\right) \min\left\{\sqrt{\frac{\ln(2)}{\lambda_z}}, \mathsf{W}\right\},$$

where $C_2 = \mathcal{O}\Big(\|u^\star\|\sqrt{\ln(1/\delta)}\Big)$.

Proof of Claim. From Lemma 8.1, we have,

$$|R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda)| \leq 2|\hat{w}_{\lambda,z}|L_{(0,1)}(\hat{u}), \text{ and } |R(w_\lambda^\star) - R^\phi(w_\lambda^\star)| \leq 2|w_{\lambda,z}^\star|L_{(0,1)}(\hat{u}).$$

Note the following relation between zero-one risk and logistic risk.

$$\frac{L_{(0,1)}(u)}{\ln(2)} \leq L(u).$$

Thus, we get

$$\begin{aligned} R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda) + R^\phi(w_\lambda^\star) - R(w_\lambda^\star) &\leq |R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda)| + |R(w_\lambda^\star) - R^\phi(w_\lambda^\star)| \\ &\leq 2L_{(0,1)}(\hat{u})(|\hat{w}_{\lambda,z}| + |w_{\lambda,z}^\star|) \leq \ln(4)L(\hat{u})(|\hat{w}_{\lambda,z}| + |w_{\lambda,z}^\star|) \end{aligned}$$

Using Lemma 6.2, with probability at least $1-\delta$, this simplifies to

$$R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda) + R^\phi(w^\star_\lambda) - R(w^\star_\lambda) \leq \ln(4)\Big(L(u^\star) + \frac{C_2}{\sqrt{m}}\Big)(|\hat{w}_{\lambda,z}| + |w^\star_{\lambda,z}|)$$

where $C_2 = \mathcal{O}\Big(\|u^\star\|\sqrt{\ln(1/\delta)}\Big)$ is a constant. From Lemma 8.2, we have

$$|w^\star_{\lambda,z}| \leq \sqrt{\frac{\ln(2)}{\lambda_z}} \quad \text{and} \quad |\hat{w}_{\lambda,z}| \leq \sqrt{\frac{\ln(2)}{\lambda_z}}.$$

From Corollary 1, we have with probability at least $1-\delta$,

$$\mathsf{W} \geq |\hat{w}_{\lambda,z}|, |w^\star_{\lambda,z}|.$$

Substituting these results, we obtain with probability at least $1-2\delta$,

$$R(\hat{w}_\lambda) - R^\phi(\hat{w}_\lambda) + R^\phi(w^\star_\lambda) - R(w^\star_\lambda) \leq \ln(4)\Big(\frac{C_2}{\sqrt{m}} + L(u^\star)\Big)\min\left\{\sqrt{\frac{\ln(2)}{\lambda_z}}, \mathsf{W}\right\}.$$

as claimed. □

We next address the sum of terms (B) and (D).

Claim 2. With probability at least $1-2\delta$,

$$R(w^\star_\lambda) - R(w^*) + R^\phi(\hat{w}_\lambda) - R^\phi(w^\star_\lambda) \leq \min\left\{\frac{\lambda_z|w^\star_z|^2}{T(v^\star) - R(w^\star)}, 1 + \frac{\lambda_z|w^\star_{\lambda,z}|^2}{T(v^\star) - R(w^\star)}\right\}(T(v^\star) - R(w^\star)) + \frac{C_1}{\sqrt{n}},$$

where $C_1 = \mathcal{O}\left(W^2 + \kappa W\sqrt{\ln(1/\delta)} + \kappa^2\ln(1/\delta)\right)$.

Proof of Claim. From the definition of $w^\star_\lambda$, we have

$$R(w^\star_\lambda) - R(w^*) \leq \lambda_c|w^*_c|^2 + \lambda_x\|w^*_x\|^2 + \lambda_z|w^*_z|^2 - \lambda_c|w^\star_{\lambda,c}|^2 - \lambda_x\|w^\star_{\lambda,x}\|^2 - \lambda_z|w^\star_{\lambda,z}|^2.$$

From Corollary 8.4.1, we have for $\lambda = \left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}, \infty\right)$

$$R(w^\star_\lambda) \leq T(v^\star) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big).$$

Therefore

$$R(w^\star_\lambda) - R(w^*) \leq \min\Bigg\{T(v^\star) - R(w^*) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big), \lambda_c|w^*_c|^2 + \lambda_x\|w^*_x\|^2 + \lambda_z|w^*_z|^2 - (\lambda_c|w^\star_{\lambda,c}|^2 + \lambda_x\|w^\star_{\lambda,x}\|^2 + \lambda_z|w^\star_{\lambda,z}|^2)\Bigg\}. \tag{38}$$

We now evaluate $R^\phi(\hat{w}_\lambda) - R^\phi(w^\star_\lambda)$. From Lemma 6.4, we get with probability at least $1-\delta$,

$$R^\phi(\hat{w}_\lambda) \leq \hat{R}^\phi(\hat{w}_\lambda) + \frac{4(1+\sqrt{\kappa^2+2}\|\hat{w}_\lambda\|)}{\sqrt{n}}\left(1 + \sqrt{\ln\left(2 + \log_2(1+\sqrt{\kappa^2+2}\|\hat{w}_\lambda\|) + \ln(1/\delta)\right)}\right).$$

Let

$$C' := 4\left(1+\sqrt{\ln\left(2+\log_2\left(1+\sqrt{\kappa^2+2}\Big(\mathsf{W}+\mathcal{O}\Big(\frac{1}{\sqrt{m}}\Big)\Big)\right)\right)}+\ln(1/\delta)\right).$$

Hence

$$R^\phi(\hat{w}_\lambda)-\hat{R}^\phi(\hat{w}_\lambda)\le\frac{C'}{\sqrt{n}}+\frac{C'\sqrt{\kappa^2+2}\|\hat{w}_\lambda\|}{\sqrt{n}}.$$

Using the inequality $ab\le\alpha a^2+\frac{b^2}{4\alpha}$ with

$$a=\|\hat{w}_\lambda\|, b=\frac{C'\sqrt{\kappa^2+2}}{\sqrt{n}},\text{ and }\alpha=\frac{\lambda_c|\hat{w}_{c,\lambda}|^2+\lambda_x|\hat{w}_{\lambda,x}|^2+\lambda_z|\hat{w}_{\lambda,z}|^2}{\|\hat{w}_\lambda\|^2}\ge\min\{\lambda_c,\lambda_x,\lambda_z\}=\frac{1}{\sqrt{n}},$$

we get

$$R^\phi(\hat{w}_\lambda)\le\hat{R}^\phi(\hat{w}_\lambda)+\lambda_c|\hat{w}_{c,\lambda}|^2+\lambda_x\|\hat{w}_{\lambda,x}\|^2+\lambda_z|\hat{w}_{\lambda,z}|^2+\frac{C'}{\sqrt{n}}+\frac{C'^2(\kappa^2+2)}{4\sqrt{n}}. \tag{39}$$

Using the optimality property of $\hat{w}_\lambda$, we have

$$\hat{R}^\phi(\hat{w}_\lambda)+\lambda_c|\hat{w}_{c,\lambda}|^2+\lambda_x\|\hat{w}_{\lambda,x}\|^2+\lambda_z|\hat{w}_{\lambda,z}|^2\le\hat{R}^\phi(w^\star_\lambda)+\lambda_c|w^\star_{\lambda,c}|^2+\lambda_x\|w^\star_{\lambda,x}\|^2+\lambda_z|w^\star_{\lambda,z}|^2,$$

which when substituted into (39) gives

$$R^\phi(\hat{w}_\lambda)\le\hat{R}^\phi(w^\star_\lambda)+\lambda_c|w^\star_{\lambda,c}|^2+\lambda_x\|w^\star_{\lambda,x}\|^2+\lambda_z|w^\star_{\lambda,z}|^2+\frac{C'}{\sqrt{n}}+\frac{C'^2\kappa^2}{4\sqrt{n}}.$$

Thus with probability at least $1-\delta$,

$$R^\phi(\hat{w}_\lambda)-R^\phi(w^\star_\lambda)\le|\hat{R}^\phi(w^\star_\lambda)-R^\phi(w^\star_\lambda)|+\lambda_c|w^\star_{\lambda,c}|^2+\lambda_x\|w^\star_{\lambda,x}\|^2+\lambda_z|w^\star_{\lambda,z}|^2+\frac{C'}{\sqrt{n}}+\frac{C'^2\kappa^2}{4\sqrt{n}}. \tag{40}$$

Using Lemma 8.3, we have

$$|\hat{R}^\phi(w^\star_\lambda)-R^\phi(w^\star_\lambda)|\le(\ln(2)+K\sqrt{\kappa^2+2}\|w^\star_\lambda\|)\sqrt{\frac{\ln(2/\delta)}{n}},$$

where $K=\max_y K_y$, and K_y is the Lipschitz constant of $\ell(\cdot,y,\cdot;\cdot)$ for fixed y. Substituting this into the (40), we get with probability at least $1-2\delta$,

$$\begin{aligned}R^\phi(\hat{w}_\lambda)-R^\phi(w^\star_\lambda)\le{}&(\ln(2)+K\sqrt{\kappa^2+2}\|w^\star_\lambda\|)\sqrt{\frac{2\ln(2/\delta)}{n}}+\lambda_c|w^\star_{\lambda,c}|^2\\&+\lambda_x\|w^\star_{\lambda,x}\|^2+\lambda_z|w^\star_{\lambda,z}|^2+\frac{C'}{\sqrt{n}}+\frac{C'^2\kappa^2}{4\sqrt{n}}.\end{aligned} \tag{41}$$

Combining the bounds for $R(w_\lambda^\star) - R(w^*)$ and $R^\phi(\hat{w}_\lambda) - R^\phi(w_\lambda^\star)$ given in (38) and (41) respectively, we get with probability at least $1 - 2\delta$,

$$\begin{aligned}
R(w_\lambda^\star) - R(w^*) + R^\phi(\hat{w}_\lambda) - R^\phi(w_\lambda^\star) \leq \min \Bigg\{ & T(v^\star) - R(w^*) + \mathcal{O}\Big(\frac{\mathsf{W}^2}{\sqrt{n}}\Big), \lambda_c |w_c^*|^2 + \lambda_x \|w_x^*\|^2 + \lambda_z |w_z^*|^2 \\
& - (\lambda_c |w_{\lambda,c}^\star|^2 + \lambda_x \|w_{\lambda,x}^\star\|^2 + \lambda_z |w_{\lambda,z}^\star|^2) \Bigg\} \\
& + (\ln(2) + \sqrt{\kappa^2 + 2} \|w_\lambda^\star\|) \sqrt{\frac{2\ln(2/\delta)}{n}} \\
& + \lambda_c |w_{\lambda,c}^\star|^2 + \lambda_x \|w_{\lambda,x}^\star\|^2 + \lambda_z |w_{\lambda,z}^\star|^2 + \frac{C'}{\sqrt{n}} + \frac{C'^2 \kappa^2}{4\sqrt{n}} \\
& \leq \min\{T(v^\star) - R(w^\star) + \lambda_z |w_{\lambda,z}^\star|^2, \lambda_z |w_z^\star|^2\} + \frac{C_1}{\sqrt{n}} \\
& = \min\Big\{\frac{\lambda_z |w_z^\star|^2}{T(v^\star) - R(w^\star)}, 1 + \frac{\lambda_z |w_{\lambda,z}^\star|^2}{T(v^\star) - R(w^\star)}\Big\}(T(v^\star) - R(w^\star)) \\
& + \frac{C_1}{\sqrt{n}}.
\end{aligned}$$

as claimed. Here $C_1 = \mathcal{O}\left(W^2 + \kappa W \sqrt{\ln(1/\delta)} + \kappa^2 \ln(1/\delta)\right)$. □

Combining the bounds from Claim 1 and Claim 2, we obtain the desired result. This completes the proof of Theorem 4.1. □

10 Proof of the lower bound

We now prove Theorem 4.2.

Proof. We establish the lower bound by constructing two distributions that are indistinguishable given the partitioned data. Let X be uniformly distributed over $\mathcal{X} = [0, 1]$. Let $p = 0.2$. Define two joint distributions on (X, Y, Z) as follows:

(I) **Distribution P_1:** Conditional on $X = x$, (Y_1, Z_1) are positively correlated such that $\mathbb{P}(Y_1 = Z_1) = 1 - p$.

	$Z_1 = -1$	$Z_1 = +1$
$Y_1 = -1$	$1 - p$	p
$Y_1 = +1$	p	$1 - p$

The Bayes optimal classifier predicts the log-odds score $\hat{y} = \alpha Z$, where $\alpha = \ln \frac{1-p}{p}$. The optimal logistic risk is the binary entropy:

$$R^* = H(p) = -p \ln(p) - (1 - p) \ln(1 - p).$$

(II) **Distribution P_2:** Conditional on $X = x$, (Y_2, Z_2) are negatively correlated such that $\mathbb{P}(Y_2 = -Z_2) = 1 - p$.

	$Z_2 = -1$	$Z_2 = +1$
$Y_2 = -1$	p	$1 - p$
$Y_2 = +1$	$1 - p$	p

Here, the optimal classifier predicts the score $\hat{y} = -\alpha Z$. The optimal logistic risk is also $R^* = H(p)$.

Observe that for both distributions, the marginals are identical:

$$\mathbb{P}(Y = y|X) = 0.5 \quad \text{and} \quad \mathbb{P}(Z = z|X) = 0.5.$$

Since the training data is partitioned into $D = \{(x_i, y_i)\}$ and $S = \{(x_j, z_j)\}$, the likelihood of the training set is identical under P_1 and P_2. Thus, no learning algorithm A can distinguish between the two cases based on the training data.

We now calculate the lower bound. Let the true distribution index k be chosen uniformly from $\{1, 2\}$. Let $\hat{w}$ be the weights learnt by the algoirthm. The expected risk over the choice of distribution is:

$$\mathbb{E}_k[R(A)] = \frac{1}{2}R_1(\hat{w}) + \frac{1}{2}R_2(\hat{w}).$$

Note that in this mixture, for every (x, z), we have $\mathbb{P}(Y = 1 \mid X = x, Z = z) = 0.5$. Thus the optimal $\hat{w}$ is 0. If $\hat{w}$ is chosen to optimize performance on one specific distribution, then its risk on the other distribution will be high. Consequently, the worst-case risk of $\hat{w}$ will be poor.

The risk of the $\hat{w} = 0$ predictor is

$$R_{\text{avg}} = -\ln(0.5) = \ln 2 \approx 0.6931.$$

The expected excess risk is therefore bounded by:

$$\mathbb{E}_k[R(A)] - R^* \geq \ln 2 - H(0.2) \approx 0.19.$$

Since the *average* excess risk is at least 0.19, the excess risk must be at least 0.19 for at least one of the two distributions. Thus, with probability at least 0.5 (over the choice of the true distribution), the excess risk is at least 0.19. □